GUARD: Guided Unlearning and Retention via Data Attribution for Large Language Models

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Abstract

Unlearning large language models (LLMs) is an important problem due to copyright and privacy concerns. A key challenge in LLM unlearning is unintended forgetting because the removal of data inadvertently impairs the utility of the model. While prior work has primarily focused on architectural innovations, the influence of data-level factors on unlearning performance remains underexplored. To address this issue, we introduce GUARD, a new framework for Guided Unlearning And Retention via Data Attribution. GUARD uses a lightweight proxy data attribution metric tailored to LLM unlearning which efficiently quantifies the "alignment" between the Forget and Retain (or synthetic supplement) Sets. This metric leads to a new unlearning objective that assigns adaptive, nonuniform unlearning weights to samples which decay exponentially with respect to their proxy attribution scores, and thereby provably mitigates unintended retention loss. Extensive experiments on the TOFU benchmark with a 10% forget set demonstrate that applying GUARD to Gradient Ascent (GA) improves the Retain Set ROUGE-L from 0.18 to 22.40 (a 124× increase) and the Retain Probability from 0.06% to 17.50%, reflecting substantial functional recovery from the GA collapse. Furthermore, GUARD combined with NPO (Negative Preference Optimization) outperforms Representation Misdirection for Unlearning (RMU) by offering 33.80% and 53.10% reductions in the Retain Set Sacrifice Rates computed from ROUGE-L and Probability (ρ_r and ρ_p), and a 16.01% improvement in knowledge retention on the MUSE NEWS Retain Set. These findings persist even when attribution scores are computed using synthetic or highly subsampled (20%) Retain Sets, where NPO+GUARD still reduces ρ_r and ρ_p on the Retain Set by up to 43.89% and 48.13% over NPO. Similar improvements are observed for MUSE, with a tolerable absolute privacy leakage degradation of 13.7 compared to the best-performing existing method in the privacy category.

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1 Introduction

Although LLMs offer state-of-the-art performance across a wide range of natural language processing tasks (Ouyang et al., 2022; Wei et al., 2021; Touvron et al., 2023a), their deployment has raised substantial privacy and security concerns. LLMs are typically trained on massive corpora of data that contain sensitive or proprietary information and this creates the risk of unintended data leakage and misuse (Carlini et al., 2021; Yao et al., 2024a). Privacy challenges have prompted a growing interest in developing *unlearning mechanisms* for the selective removal of specific information from LLMs without the use of retraining (Yao et al., 2024b). Despite significant advances in unlearning (Liu et al., 2025; Zhang et al., 2024a; Wang et al., 2024; Jia et al., 2024; Wei et al., 2025), fine-tuning-based LLM unlearning methods still suffer from *excessive forgetting* and *retention degradation* (Ha et al., 2025), compromising the utility of the retained knowledge. Similarly, inference-time LLM unlearning models (Liu et al., 2024a; Ji et al., 2024), which avoid updates and are hence computationally attractive, do not guarantee forgetting as sensitive information remains embedded within the unaltered model parameters (Appendix A.1). For a more in-depth treatment of prior work on unlearning and attribution, we refer the reader to Appendix A. Current limitations of LLM unlearning methods consequently include one or more of the following categories: **a)** *degraded retention*, owing to unlearning performance losses on the Retain Set (Zhang et al., 2024a); **b)** *privacy leakage*, caused by unacceptable partial removal of sensitive information (Liu et al., 2024a); **c)** *computational inefficiency,* due to the frequent requirement for auxiliary LLM guidance/evaluation (Ji et al., 2024); **d)** *lack of theoretical performance guarantees.* To address the above problems, we propose a new unlearning method that ensures *utility preservation, effective forgetting, computational efficiency, and comes with formal theoretical guarantees,* termed GUARD (**G**uided **U**nlearning **A**nd **R**etention via **D**ata Attribution). GUARD has the following important properties:

a) Retention and b) Privacy Loss of GUARD. Most existing unlearning methods use unprincipled allocation of model unlearning weights across forget samples despite the fact that neglecting to account for the impact of a sample can lead to utility loss. To counter this issue, GUARD *dynamically adjusts unlearning weights* based on the influence of each forget sample on the performance over the retained data. More precisely, it allocates unlearning weights so that the data-wise loss weight decreases with the sample-wise contribution to retention, reducing the likelihood of unintentionally degrading retained knowledge. To this end it uses a *specialized* gradient-based retention attribution score that quantifies the "alignment" of a forget sample with the retained knowledge, and is tailored to unlearning without requiring retraining or computation of the Hessian. To ensure effective forgetting, GUARD also includes a temperature-controlled reverse unification mechanism that regulates the variance of unlearning weights across samples. This enables the method to mitigate retention degradation by down-weighting high-retention-impact samples and maintain effective forgetting.

c) Computational Efficiency of GUARD. Since GUARD has to perform computationally efficient data attribution on LLMs, one has to reconsider existing attribution methods. Conventional data attribution methods estimate the influence of individual training samples on overall model performance by comparing model outputs with and without the inclusion of each sample (Koh & Liang, 2017). These approaches typically require model retraining or Hessian-based approximations, making them computationally prohibitive for LLMs. Moreover, evaluating the influence of samples based on global performance contradicts the unlearning objective which aims to *decrease* the performance on the Forget Set and simultaneously *preserve* it on the Retain Set (see Appendix A.2). Our solution to this problem is a novel attribution score tailored to LLM unlearning. Inspired by gradient-based sample-wise representations (Nguyen et al., 2024; Wang et al., 2025b), we define the *retention attribution score* of a forget sample as the inner product between its gradient and the gradient averaged over all retain samples. The score captures the alignment between the "knowledge" encoded by a forget sample and that of the retain distribution. It also avoids retraining or complicated Hessian estimation inherent to prior attribution techniques. Furthermore, through explicit separation of forget and retain knowledge before inter-sample influence computations, the score effectively captures the contribution of a sample to both utility and unlearning.

d) Theoretical Guarantees of GUARD. We also provide a rigorous theoretical analysis demonstrating that, compared to conventional unlearning baselines such as Gradient Ascent (GA), GUARD has the following advantages: (1) the influence of a forget sample on the Retain Set, as estimated by conventional Hessian-based attribution methods, equals a global rescaling of its retention attribution score plus a residual bounded by the spectral spread of the Hessian, and the rescaling is absorbed by the temperature; (2) it effectively reduces

the loss on the Retain Set, leading to enhanced *absolute retention*; (3) it maintains comparable degradation on the Forget Set, ensuring strong *absolute forgetting*; and (4) it reduces the retention-performance tradeoff over baselines under an explicit sufficient condition, thereby improving relative retention efficiency (with experimentally confirmed results).

Empirical Validation. To test the practical performance of GUARD, we conduct extensive experiments on the TOFU (Maini et al., 2024) and MUSE (Shi et al., 2024) benchmarks, both specifically designed for LLM unlearning. *All TOFU experiments strictly follow the official TOFU protocol* (AdamW, lr 10^{-5}, weight decay 0.01, 5 unlearning epochs, effective batch size 32, linear warm-up in epoch 1), and we report numbers at the final (5-th) unlearning epoch, matching the convention adopted by the TOFU leaderboard. We cover four non-loss reweighting baselines (GA, GD, KM, PO) and their GUARD-augmented counterparts (GA+GUARD, GD+GUARD, KM+GUARD, PO+GUARD); four loss reweighting baselines (NPO, SimNPO, FLAT, SatImp) and NPO+GUARD (NPO was chosen for integration with GUARD due to its stability and performance); and RMU, a stable unlearning method frequently used in practice. NPO+GUARD outperforms RMU in retention for both datasets: It reduces the Retain Set Sacrifice Rates computed from ROUGE-L and Probability (ρ_r and ρ_p) on TOFU by 33.80% and 53.10% over RMU when forgetting 10% of the training data; on the MUSE NEWS Retain Set, it enhances knowledge retention by 16.01% over RMU (Retain KnowMem 50.0 vs. 43.1). Furthermore, GUARD remains highly effective even with limited Retain Set access (or synthetic data replacements): When using only 20% of the Retain Set, NPO+GUARD attains Retain Set $\rho_r = 17.78$ and $\rho_p = 5.69$, substantially better than the NPO baseline values of 31.69 and 10.97 and approaching the values 14.32 and 2.87 achieved by NPO+GUARD on the full Retain Set. Table 1 summarizes how GUARD differs from representative prior methods.

2 LLM Unlearning Preliminaries

We start with a description of the standard setup for unlearning of LLMs (Feng et al., 2024; Maini et al., 2024; Zhang et al., 2024a; Wang et al., 2024; Jia et al., 2024; Qiu et al., 2025; Liu et al., 2024a; Ji et al., 2024; Yao et al., 2024b; Zhang et al., 2024b).

Data - The Forget and Retain Sets. We define a data point as (x, y), where $x \in \mathbb{R}^p$ represents the input of the LLM, while $y \in \mathbb{R}^q$ denotes the corresponding ground-truth label or response. The whole data set used for fine-tuning an LLM is denoted by $D_0 = \{(x_i, y_i)\}_{i\in[n_0]}$, where n_0 is the total number of data points. The Forget Set, denoted by D_f, consists of data points requested for removal, while the Retain Set, denoted by D_r, contains data points to be kept. The index sets of these data points are defined as $I_f := \{i \mid (x_i, y_i) \in D_f\}$ and $I_r := \{j \mid (x_j, y_j) \in D_r\}$, respectively. We also let $n_f := |I_f|$ and $n_r := |I_r|$ represent the respective cardinalities of the sets. Clearly, $n_f + n_r = n_0$ and $D_0 = D_f \cup D_r$.

The Empirical Loss. Let Θ be the parameter space, and let the LLM be parameterized by $\theta \in \Theta$. Given a loss function $l : \mathbb{R}^p \times \mathbb{R}^q \times \Theta \to \mathbb{R}$, the loss incurred on a data point (x, y) equals $l(x, y; \theta)$. For a data set D, the empirical loss is given by $L_D(\theta) = \frac{1}{|D|} \sum_{(x_i,y_i)\in D} l(x_i, y_i; \theta)$.

Standard LLM Unlearning. Henceforth, *standard LLM unlearning* refers to the process of adapting an LLM initially trained on D_0 in order to remove the influence of a designated Forget Set D_f. One typical approach is Gradient Ascent (GA), which updates the model parameters θ_0 as follows:

$$\theta_{GA} = \theta_0 + \eta \cdot \nabla_\theta L_{D_f}(\theta_0) = \theta_0 + \eta \cdot \sum_{i\in I_f} \frac{1}{n_f} \cdot \nabla_\theta l(x_i, y_i; \theta_0), \tag{1}$$

where η denotes the unlearning rate. Alternative unlearning methods are discussed in Appendix D.1.

3 GUARD

The Pipeline. Unlike traditional methods that assign retention-unaware unlearning weights to all data points (e.g., uniform weights in GA, equal to $\frac{1}{n_f}$, see Eq. (1)), GUARD introduces a new objective that controllably leverages sample loss weights that are inversely correlated to data attribution scores. Additionally, GUARD uses a new data attribution scheme specifically tailored to the LLM unlearning setting. The GUARD pipeline comprises the steps shown in Fig. 1:

Table 1: Positioning of GUARD. **Granularity** is the unit over which unlearning strength may vary ("—" = uniform). **Retain set**: × = unused; *reg.* = used only as an additive regularizer, which leaves the allocation across forget units unchanged; ✓ = determines the weights themselves. **Hessian-free** = no second-order estimate and no retraining, as originally defined; in Section 5 we compare against the *first-order* approximations of RelatIF and RIF so the comparison isolates the score, not the curvature machinery. **Guar.** = formal bound on retain-side loss relative to a baseline. References: NPO (Zhang et al., 2024a), SimNPO (Fan et al., 2024), FLAT (Wang et al., 2024), SatImp (Yang et al., 2025), WGA and TNPO (Wang et al., 2025b), WAGLE (Jia et al., 2024), influence functions (Koh & Liang, 2017), TRAK (Park et al., 2023), EK-FAC (Grosse et al., 2023), RelatIF (Barshan et al., 2020), and RIF (Rubinstein & Hopkins, 2025).

Method	**Signal setting unlearning strength**	**Granularity**	**Retain set**	**Hess.-free**	**Guarantee**
GA	uniform	—	×	✓	×
PO, GD, KM, RMU	uniform (plus a retain regularizer)	—	*reg.*	✓	×
NPO, SimNPO	forget likelihood (ratio / reference-free)	per-sample	*reg.*	✓	×
FLAT	f-divergence, template vs. forget response	per-branch	×	✓	×
SatImp	token saturation / importance	per-token	*reg.*	✓	×
WGA, TNPO	forget token likelihood (G-effect motivated)	per-token	×	✓	×
WAGLE	per-weight product of forget and retain gradients	per-weight	✓	✓[†]	×
Influence fn., TRAK, EK-FAC	global loss influence weighted by the inverse Hessian	per-sample	×	×	×
RelatIF, RIF	influence normalized by the sample's self-influence, or rescaled by its leverage score	per-sample	×	×	×
GUARD (ours)	alignment of each forget gradient with the average retain gradient	per-forget-sample	✓	✓	✓

[†] WAGLE's score is derived under a *diagonal* Hessian assumption, so only first-order gradients are computed in practice.

Step 1. *Computing the data attribution scores for retention:* Estimating proxy data attribution scores for all data points in the Forget Set.
Step 2. *Deriving retention-aware unlearning weights:* Calculating unlearning weights that are inversely related to the computed attribution scores, with the degree of adjustment modulated by a temperature parameter τ.
Step 3. *Performing GUARD unlearning*: Optimizing the GUARD objective using the retention-aware unlearning weights.

Data Attribution for LLM Unlearning. Conventional data attribution methods estimate the influence of individual training samples by measuring the change in the model outputs when a sample is removed from the training set. These approaches typically rely on retraining or Hessian-based computations (see Appendix A.2). However, such methods are impractical in the context of LLM unlearning for two key reasons: (1) their computational cost is prohibitive at the operational scale of LLMs; and (2) the objective of unlearning differs fundamentally from standard attribution, i.e., effective unlearning necessitates different performance "shifts", namely, degraded performance on the Forget Set D_f and preserved performance on the Retain Set D_r (rather than a uniform unidirectional performance change over the entire dataset). To address these challenges, we propose a proxy attribution score a_i^{GUARD} for $i \in I_f$ as

$$a_i^{\mathrm{GUARD}} := J_{\mathrm{avg}}^{D_r}(\theta_0)^\top J_i(\theta_0), \quad \forall i \in I_f. \tag{2}$$

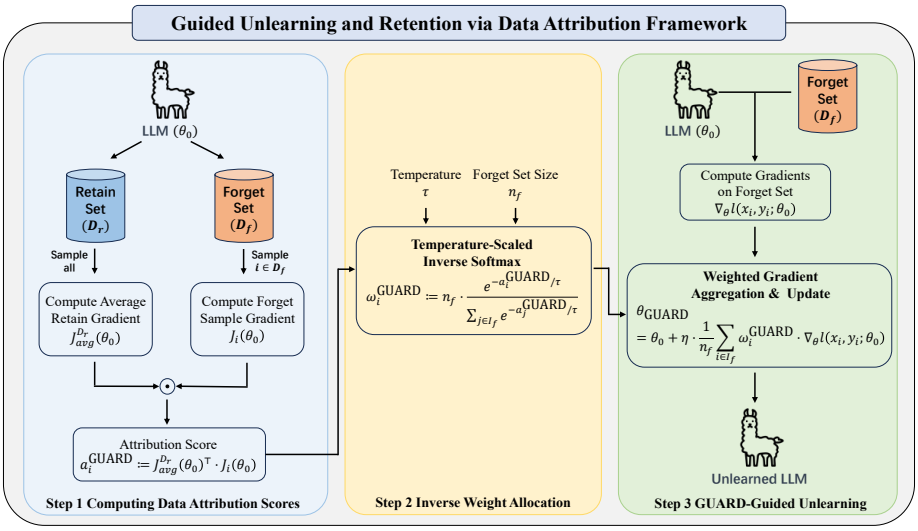


Figure 1: The pipeline. **Step 1:** Compute the proxy data attribution score a_i^{GUARD} for each forget sample as the inner product of its gradient and the average gradient of the Retain Set. **Step 2:** Convert the scores into per-sample unlearning weights ω_i^{GUARD} via temperature-scaled reverse Softmax unification. **Step 3:** Apply the retention-aware weights to the unlearning objective.

Here, $J^{D_r}_{avg}(\theta_0)$ denotes the average gradient over D_r and $J_i(\theta_0)$ is the gradient of sample i, both evaluated for the initial model parameters θ_0. This formulation captures the alignment between the information content of the Forget and the Retain Set. Representing sample-level influence via gradients is a well-established practice, particularly for LLMs (Nguyen et al., 2024; Wang et al., 2025b). Note that compared to other first-order influence measures not specifically tailored to unlearning such as RelatIF, the attribution score a_i^{GUARD} defined in (2) preserves gradient magnitude. For unlearning, *this magnitude is informative,* as it typically reflects the "difficulty" of a sample or the strength of its encoded knowledge (experimental comparison with alternative influence functions can be found in Appendix E.9). Beyond its intuitive interpretation and computational efficiency, our proxy score also admits theoretical analysis (Appendix B) and exhibits robustness to the choice of the Retain Set (Section 5 and Appendix E.6).

Retention-Aware Unlearning Weights. A large retention attribution score for a forget sample i indicates that the sample (x_i, y_i) has a strong positive influence on the performance of the model over the Retain Set D_r (e.g., its removal leads to a large performance drop on D_r). We assign unlearning weights to data samples that decrease exponentially in their attribution scores; at the same time, we modulate the variance of the weight distribution by a temperature-controlled softmax. We define the unlearning weight for $i \in I_f$ as

$$\omega_i^{\text{GUARD}} := n_f \cdot \frac{e^{-a_i^{\text{GUARD}}/\tau}}{\sum_{j\in I_f} e^{-a_j^{\text{GUARD}}/\tau}}, \tag{3}$$

where as before, a_i^{GUARD} is the attribution score defined in Eq. (2), τ is the temperature hyperparameter, and $n_f = |I_f|$ denotes the size of the Forget Set. We refer to the mapping in Eq. (3) from scores to weights as *reverse Softmax unification.* The unlearning weights ω_i^{GUARD} satisfy the:

1. Normalization Property, which asserts that $\frac{1}{n_f}\sum_{i\in I_f} \omega_i^{\text{GUARD}} = 1$.

2. Inverse Attribution Score Property, which asserts that unlearning weights should be decreasing in the attribution scores of the samples; this mitigates unintended loss of knowledge in D_r during unlearning of D_f.

3. Controlled Variance Property, which asserts that the degree of weight differentiation is governed by a temperature parameter τ so as to enable effective forgetting on D_f. Comparable forgetting to baselines is supported theoretically (Section 4) and validated empirically (Section 5).

Note that the inner product a_i^{GUARD} in Eq. (2) quantifies how much a forget sample contributes to updates in the direction of the retain gradient. Samples whose gradients are strongly aligned with the average

Retain Set gradient $J_{\text{avg}}^{D_r}(\theta_0)$ cause updates that degrade retention and should therefore be down-weighted during unlearning. The softmax weighting in Eq. (3) implements this idea by assigning smaller weights to high-alignment scores, while preserving normalization across the Forget Set. The temperature τ controls the strength of suppression. *This perspective naturally connects to a constrained unlearning formulation that maximizes the loss on the Forget Set while limiting degradation on the Retain Set.* A formal connection to a constrained optimization view is provided in Section 4.

Unlearning with GUARD. For a given Forget Set D_f and a model parameterized by θ, our new GUARD unlearning loss objective, $L^{\text{GUARD}} : \Theta \to \mathbb{R}$, is defined as

$$L_{D_f}^{\text{GUARD}}(\theta) := \frac{1}{n_f} \sum_{i \in I_f} \omega_i^{\text{GUARD}} \cdot l(x_i, y_i; \theta),$$

where ω_i is the unlearning weight defined in (3). Using the GA framework from Eq. (1) as an example, the model update rule under GUARD becomes:

$$\theta_{\text{GA}}^{\text{GUARD}} = \theta_0 + \frac{\eta}{n_f} \sum_{i \in I_f} \omega_i^{\text{GUARD}} \cdot \nabla_\theta \left[l(x_i, y_i; \theta_0) \right]. \tag{4}$$

As previously discussed, such an unlearning weight allocation (Eq. (4)) mitigates unintended forgetting of retained knowledge while it also ensures effective forgetting of undesired information (GUARD objectives for methods other than GA are described in Appendix D.2).

The GUARD Algorithm. The above approaches are summarized in Algorithm 1, which outlines the implementation of GUARD within the GA unlearning framework (Eq. (1)). The algorithm is presented as a single update step for clarity. In our experiments, we follow the official TOFU multi-epoch protocol (5 unlearning epochs) (Maini et al., 2024): the per-sample attribution-driven weights ω_i^{GUARD} are computed *once* on the fine-tuned model checkpoint and reused across all unlearning epochs, so the additional cost over the underlying baseline is a one-time preprocessing step.

Algorithm 1 GUARD: Guided Unlearning and Retention via Data Attribution

1: **Input**: Initial model weights θ_0; unlearning rate η; temperature τ.
2: Compute retention attribution a_i^{GUARD} for all $i \in I_f$ using Eq. (2) {Prepare Attribution}
3: Compute unlearning weights ω_i for all $i \in I_f$ using Eq. (3) {Prepare Sample-wise Weights}
4: Update model θ_0 to $\theta_{\text{GA}}^{\text{GUARD}}$ by Eq. (4) {Update Model}
5: **Output**: Unlearned model weights $\theta_{\text{GA}}^{\text{GUARD}}$.

4 Theoretical Analysis

Let $\psi \in \{0, \text{GA}, \text{GUARD}\}$ denote the superscript corresponding to a model trained without unlearning, trained with baseline unlearning (GA), and GUARD, respectively. For a model parameterized by θ^ψ, we define the empirical loss over the Forget Set D_f and the Retain Set D_r as:

$$L_{D_f}^{\psi} = \frac{1}{n_f} \sum_{i \in I_f} l(x_i, y_i; \theta^\psi), \quad L_{D_r}^{\psi} = \frac{1}{n_r} \sum_{i \in I_r} l(x_i, y_i; \theta^\psi), \tag{5}$$

where $n_f = |I_f|$, $n_r = |I_r|$. Let $g_i := \nabla_\theta l(x_i, y_i; \theta^0)$ denote the gradient of sample i at the model before unlearning, $\theta^0 = \theta_0$. Let $\overline{g}_f := \frac{1}{n_f} \sum_{i \in I_f} g_i$ and $\overline{g}_r := \frac{1}{n_r} \sum_{i \in I_r} g_i$ denote the average gradients over the sets D_f and D_r, respectively. We denote the global alignment (score) between the two sets as $\kappa := \langle \overline{g}_f, \overline{g}_r \rangle$ and the normalized alignment $\delta_\kappa := \kappa / \tau$, where τ denotes the GUARD temperature. For each forget index $j \in I_f$, the per-sample alignment is defined as $a_j := \langle \overline{g}_r, g_j \rangle$. We further let $\delta_a := \max_{j \in I_f} |a_j| / \tau$ denote the largest normalized per-sample alignment and $G := \max_{i \in I_f \cup I_r} \|g_i\|_2$ the largest gradient norm. We compare a single unlearning step of GA and GUARD from θ^0, i.e., $\theta^{\text{GA}} := \theta^0 + \eta\, \overline{g}_f$ and $\theta^{\text{GUARD}} := \theta^0 + \eta\, \overline{g}'_f$, as in Eq. (1) and Eq. (4), where $\overline{g}'_f := \frac{1}{n_f} \sum_{i \in I_f} \omega_i^{\text{GUARD}} g_i$ is the reweighted average forget gradient. We also define the

alignment variance

$$\sigma_\kappa^2 := \frac{1}{n_f}\sum_{j\in I_f} a_j^2 - \Big(\frac{1}{n_f}\sum_{j\in I_f} a_j\Big)^2 \geq 0. \tag{6}$$

Assumption 1. *The global alignment κ and the per-sample alignments a_j, $j \in I_f$, satisfy: (1) $\kappa > 0$; (2) $|a_j|/\tau \ll 1$, $\forall\, j \in I_f$.*

Here, (1) reflects feature-sharing of LLMs: Without positive alignment, a GA step on D_f would not increase the loss on D_r to first order; and (2) reflects weak per-sample alignment, enabling selective unlearning. Assumption 1 implies $0 < \delta_\kappa \ll 1$.

Assumption 2. *The Hessians of L_{D_f} and L_{D_r} are bounded near θ^0, and η is small enough that the second-order Taylor remainders, which are of order $\eta^2 G^2$, are negligible.*

This is a standard influence assumption (Koh & Liang, 2017), which is also valid for small unlearning perturbations.

Let the centered empirical forget gradient covariance be $\mathrm{Cov}_f := \frac{1}{n_f}\sum_{j\in I_f}(g_j - \bar{g}_f)(g_j - \bar{g}_f)^\top$ and the uncentered empirical forget gradient one $\Sigma_f := \frac{1}{n_f}\sum_{j\in I_f} g_j g_j^\top = \bar{g}_f\bar{g}_f^\top + \mathrm{Cov}_f$. We next study the performance of GUARD on the Retain Set (Theorem 1), the Forget Set (Theorem 2), and ultimately, on the newly defined Sacrifice Rate (Theorem 3). The proofs are deferred to Appendix C.

Theorem 1. *Under Assumptions 1 and 2, the losses on the Retain Set of* GUARD *and GA satisfy*

$$L_{D_r}^{GA} - L_{D_r}^{GUARD} = \frac{\eta}{(1-\delta_\kappa)\tau}\cdot\sigma_\kappa^2 \;+\; \eta\cdot O(\tau\,\delta_a^3) \;+\; O(\eta^2 G^2).$$

Under Assumption 1, we have $\delta_\kappa \ll 1$. Hence, to first order, GUARD performs better on the Retain Set than GA. The improvement scales linearly with the unlearning rate η. However, excessively large η's may violate the small-update assumption and degrade the accuracy of the first-order approximation. Furthermore, the retention gain is proportional to the alignment variance σ_κ^2: A large variance indicates that forget samples differ strongly in how much they align with the Retain Set, which is the heterogeneity GUARD exploits. Note that when $a_j = \kappa$, $\forall j \in I_f$, that is, every forget gradient has the same alignment with the retain gradient, $\sigma_\kappa = 0$. Hence, $\omega_j^{\mathrm{GUARD}} = 1$, and the weights become uniform. We now characterize the Forget Loss of GUARD.

Theorem 2. *Under Assumptions 1 and 2, we have*

$$\Big|L_{D_f}^{GA} - L_{D_f}^{GUARD}\Big| \leq \frac{\eta\sigma_\kappa}{\tau-\kappa}\sqrt{\lambda_{\max}(\mathrm{Cov}_f)}\|\bar{g}_f\|_2 \;+\; \eta\cdot O(\delta_a^2 G^2) \;+\; O(\eta^2 G^2),$$

where all the parameters are as previously defined.

Next, we define the *Sacrifice Rate* $\rho_\epsilon(D)$ which measures the relative performance degradation on a data set D other than the Forget Set per unit of performance decrease on the Forget Set: The lower the rate, the better the relative knowledge retention (see also Section 5 for additional absolute metrics):

$$\rho_\epsilon(D) = \frac{\epsilon(D;\theta^0) - \epsilon(D;\theta)}{\epsilon(D_f;\theta^0) - \epsilon(D_f;\theta)}. \tag{7}$$

A negative Sacrifice Rate indicates that the performance on D improved during unlearning. In our theoretical analysis, we take ϵ to be the loss and $D = D_r$, i.e., $\rho^\psi := (L_{D_r}^\psi - L_{D_r}^0)/(L_{D_f}^\psi - L_{D_f}^0)$ for $\psi \in \{\mathrm{GA}, \mathrm{GUARD}\}$.

For the Sacrifice Rate metric, we have the following result for $\sigma_\kappa > 0$.

Theorem 3. *Let Assumptions 1 and 2 hold true. If, in addition,*

$$\|\bar{g}_f\|_2 > \max\left\{\frac{\sigma_\kappa}{\tau-\kappa}, \frac{\kappa}{\sigma_\kappa}\right\}\sqrt{\lambda_{\max}(\mathrm{Cov}_f)}, \tag{8}$$

then, to leading order (i.e., ignoring the remainders in Theorems 1 and 2), GUARD achieves a Sacrifice Rate no worse than that of GA, i.e., $\rho^{GA} - \rho^{GUARD} \geq 0$.

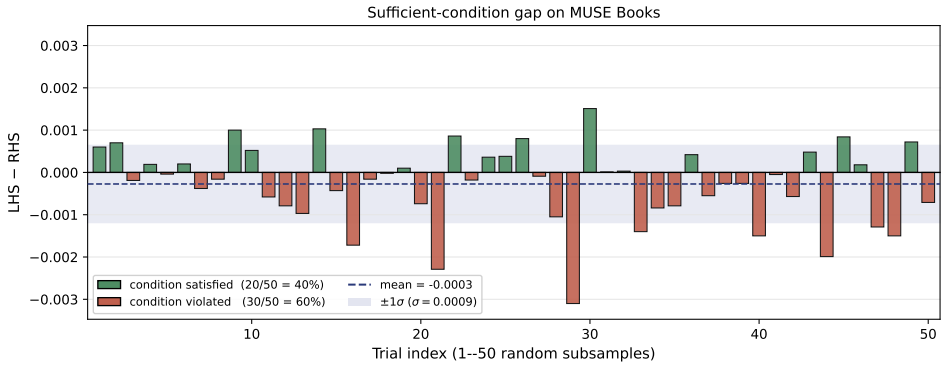


Figure 2: Empirical distribution of LHS − RHS in the sufficient condition (8) across 50 trials, each on a random subsample of MUSE BOOKS ($\tau = 0.03$). Green bars indicate trials where the sufficient condition is satisfied, red bars indicate violations. The condition holds for 20 of the 50 trials, and the gaps are small ($|\text{LHS} - \text{RHS}| < 0.004$).

GUARD performs better than GA when the norm of the mean forget gradient $\|\bar{g}_f\|_2$ is large relative to the square root of the largest eigenvalue of the forget gradient covariance matrix, $\sqrt{\lambda_{\max}(\text{Cov}_f)}$. We further observe that the alignment variance σ_κ^2, which captures the variability of the per-sample alignment a_i for $i \in I_f$, plays a critical role: A larger σ_κ^2 reflects greater heterogeneity across samples. In this case, the weighting mechanism in GUARD effectively separates high- and low-alignment samples by down-weighting samples that most increase the Retain Loss and up-weighting less harmful ones.

Consider a spiked covariance model with $\text{Cov}_f = \nu^2 I + U\Lambda U^\top$, with $U^\top U = I_k$, $\Lambda \succeq 0$, and let $\cos^2\phi := \|U^\top \bar{g}_f\|_2^2 / \|\bar{g}_f\|_2^2$ be the squared cosine of the angle between $\bar{g}_f$ and the spike subspace $\text{span}(U)$. Under this model, the second condition in (8), namely, $\|\bar{g}_f\|_2 \geq (\kappa/\sigma_\kappa)\sqrt{\lambda_{\max}(\text{Cov}_f)}$, can be relaxed to

$$\|\bar{g}_f\|_2 \geq \frac{\kappa}{\sigma_\kappa}\sqrt{\nu^2 + \lambda_{\max}(\Lambda)\cos^2\phi}. \tag{9}$$

Here, κ/σ_κ is the mean of the per-sample alignments a_j divided by the standard deviation σ_κ, and it is small when forget samples differ strongly in how much they align with the Retain Set, which is the exact type of heterogeneity GUARD exploits. Also, $\sqrt{\nu^2 + \lambda_{\max}(\Lambda)\cos^2\phi}$ upper-bounds the standard deviation of the per-sample forget gradients projected onto the direction of $\bar{g}_f$. Condition (9) therefore holds when the mean forget gradient is large compared with how much individual forget gradients vary along it and when the alignments are heterogeneous. If $\bar{g}_f \perp \text{span}(U)$, i.e., $\cos\phi = 0$, the spike drops out entirely, and $\sqrt{\nu^2 + \lambda_{\max}(\Lambda)\cos^2\phi} = \nu$. If $\bar{g}_f \in \text{span}(U)$, i.e., $\cos\phi = 1$, $\sqrt{\nu^2 + \lambda_{\max}(\Lambda)\cos^2\phi} = \sqrt{\lambda_{\max}(\text{Cov}_f)}$, and $\|\bar{g}_f\|_2 \geq (\kappa/\sigma_\kappa)\sqrt{\lambda_{\max}(\text{Cov}_f)}$ in (8) follows. Thus, $\sqrt{\nu^2 + \lambda_{\max}(\Lambda)\cos^2\phi}$ increases with $\cos\phi$, from ν when $\bar{g}_f \perp \text{span}(U)$ to $\sqrt{\lambda_{\max}(\text{Cov}_f)}$ when $\bar{g}_f \in \text{span}(U)$, where (9) coincides with (8). Hence, the condition is easiest to satisfy when the dominant fluctuations of the per-sample forget gradients are not along their mean, and it becomes the original condition (8) when the two directions align.

The Special Case of Isotropic Gradients. Setting $\Lambda = 0$ in the spiked model gives isotropic gradients, i.e., $\text{Cov}_f = \nu^2 I$ for some $\nu > 0$. Gradients of LLMs are not isotropic, but this special case makes the results above even easier to interpret. Since $\sigma_\kappa^2 = \bar{g}_r^\top \text{Cov}_f\, \bar{g}_r = \nu^2\|\bar{g}_r\|_2^2$, Theorem 1 shows that, to first order, GUARD reduces the loss on the Retain Set by $\eta\,\nu^2\|\bar{g}_r\|_2^2/(\tau - \kappa)$ compared to GA. Moreover, the second condition in (8) is no longer needed, and the leading-order term of $\rho^{\text{GA}} - \rho^{\text{GUARD}}$ is nonnegative whenever $\|\bar{g}_f\|_2^2 > \nu^2\kappa/(\tau - \kappa)$ (Appendix C).

Note that Theorem 3 provides a *conservative sufficient condition*; for the sake of completeness, we also perform empirical evaluations of the condition as outlined below.

Practical Validity of the Sufficient Condition. All quantities in condition (8) are empirical gradient statistics, so the condition can be checked directly. On the BOOKS corpus of the MUSE benchmark, we check it at the released target model with the temperature $\tau = 0.03$ used throughout the paper, in 50 trials, each on a random subsample, following the estimation protocol in Appendix C.4. As shown in Figure 2, the

condition is satisfied in 40% of the trials, and the violations are small. On TOFU with a 10% forget split, the same check succeeds in 28% of the trials (Appendix C.4), since the gradient statistics in (8) depend on the data set and the model. Even when the condition is violated, GUARD still improves retention over GA on both benchmarks (Section 5 and Table 12). These experiments use multiple AdamW steps rather than the single gradient step analyzed above, so they illustrate rather than test the theorem.

Lagrangian Multiplier Interpretation of GUARD. To formally ground the rationale of the weighting mechanism of GUARD, we analyze a linearized unlearning objective subject to a hard retention constraint.

Let $\theta \in \mathbb{R}^d$ denote the model parameters. Standard unlearning (e.g., GA) aims to find a parameter update Δ that maximizes the loss increase on the Forget Set. However, an ideal unlearning algorithm should do so while ensuring the loss increase on the Retain Set remains bounded by some ε.

First-Order Constrained Optimization. Using a first-order Taylor expansion, the above problem can be formulated as a linear objective with a quadratic regularizer:

$$\max_{\Delta} \quad \overline{g}_f^\top \Delta - \frac{\zeta}{2}\|\Delta\|^2 \qquad \text{s.t.} \quad \overline{g}_r^\top \Delta \le \varepsilon, \tag{10}$$

where $\overline{g}_f$ and $\overline{g}_r$ are the average gradients over the Forget and Retain Sets as defined above, and $\zeta > 0$ is a regularization parameter controlling the step size (trust region).

Lagrangian Dual Solution. One can form the Lagrangian for Eq. (10) as follows:

$$\mathcal{L}(\Delta, \lambda) = -\overline{g}_f^\top \Delta + \frac{\zeta}{2}\|\Delta\|^2 + \lambda(\overline{g}_r^\top \Delta - \varepsilon),$$

where $\lambda \ge 0$ denotes the Lagrange multiplier. The Karush-Kuhn-Tucker (KKT) optimality conditions with respect to Δ yield the stationarity equation:

$$\nabla_\Delta \mathcal{L} = -\overline{g}_f + \zeta\Delta^\star + \lambda\overline{g}_r = 0 \quad \Longrightarrow \quad \Delta^\star = \frac{1}{\zeta}(\overline{g}_f - \lambda\overline{g}_r).$$

If the constraint is active (i.e., the unlearning update threatens retention utility), substituting $\Delta^\star$ into the binding constraint $\overline{g}_r^\top \Delta^\star = \varepsilon$ allows us to solve for the optimal multiplier $\lambda^\star$:

$$\overline{g}_r^\top \left[\frac{1}{\zeta}(\overline{g}_f - \lambda^\star \overline{g}_r)\right] = \varepsilon \quad \Longrightarrow \quad \lambda^\star = \frac{\overline{g}_r^\top \overline{g}_f - \zeta\varepsilon}{\|\overline{g}_r\|^2}. \tag{11}$$

Crucially, the optimal multiplier $\lambda^\star$ grows with the inner product $\langle \overline{g}_r, \overline{g}_f \rangle$, which represents the global alignment (or conflict) between the forget and retain gradients. Substituting $\lambda^\star$ back into the expression for $\Delta^\star$, the optimal update direction becomes:

$$\Delta^\star \propto \underbrace{\overline{g}_f}_{\text{Standard Ascent}} - \underbrace{\frac{\langle \overline{g}_r, \overline{g}_f \rangle - C}{\|\overline{g}_r\|^2}\overline{g}_r}_{\text{Retention Correction}}, \tag{12}$$

where $C = \zeta\varepsilon$ is a constant. Eq. (12) reveals that theoretically optimal unlearning requires subtracting the component of the forget gradient that projects onto the direction of $\overline{g}_r$.

Connection to GUARD. GUARD translates this idea of global subtraction into a sample-wise reweighting mechanism. Instead of computing a single global correction coefficient based on $\langle \overline{g}_r, \overline{g}_f \rangle$, GUARD assesses the conflict for each individual forget sample i, i.e.

$$a_i^{\text{GUARD}} = \langle \overline{g}_r, g_i \rangle,$$

where $g_i := J_i(\theta_0)$ denotes the gradient of forget sample i. By assigning lower weights ω_i^{GUARD} to samples with high attribution scores a_i^{GUARD}, GUARD effectively suppresses the contribution of high-conflict samples to the aggregated update $\overline{g}'_f$. This acts as a *soft realization* of the projection in Eq. (12): Rather than explicitly subtracting the retain gradient vector, we dampen the source of the conflict itself (the specific samples aligned with $\overline{g}_r$).

5 Experiments

Table 2: Unlearning experiments for LLaMA-2-7B on 10% of TOFU data points. Arrows (↑/↓) indicate whether higher/lower values are preferred; bold numbers indicate the lower value within each (baseline, baseline+GUARD) pair for GA, KM, PO, and GD, and the lowest value among the loss-reweighting methods, RMU, and NPO+GUARD. GUARD reduces retention degradation compared to baselines.

Methods	Retain Set(↓)			Real Author Set(↓)			Real World Set(↓)		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
GA	100.17	99.87	121.09	94.80	21.71	69.67	89.72	17.63	46.36
GA + GUARD	**78.98**	**83.46**	**84.29**	**79.63**	**16.45**	**49.90**	**68.39**	**12.05**	**26.64**
KM	100.05	99.86	118.67	94.71	17.92	60.88	89.64	16.72	45.51
KM + GUARD	**82.13**	**86.05**	**89.15**	**81.74**	**13.75**	**42.74**	**70.21**	**12.84**	**29.48**
PO	21.84	36.72	60.00	45.02	34.20	317.41	4.23	17.35	198.15
PO + GUARD	**14.91**	**9.95**	**7.53**	**42.19**	**26.22**	**102.47**	**1.01**	**2.17**	**55.06**
GD	52.41	44.10	**-4.94**	12.33	**-10.41**	**-43.01**	**-0.20**	**-5.04**	**-21.37**
GD + GUARD	**39.21**	**31.87**	-1.15	**9.02**	-4.70	-19.72	-0.05	-1.66	-4.51
NPO	31.69	10.97	**-14.84**	23.36	11.64	-21.15	5.92	5.05	-16.56
SimNPO	28.82	9.77	-12.18	22.12	10.60	-22.68	5.41	4.61	-16.90
FLAT (TV)	36.93	13.12	-12.80	26.32	13.12	-18.39	6.75	6.09	-14.52
SatImp	32.86	11.72	-13.56	23.32	11.89	**-23.11**	5.95	4.69	**-18.52**
RMU	21.63	6.12	-12.64	16.60	9.87	-17.98	4.69	4.18	-13.95
NPO + GUARD	**14.32**	**2.87**	-6.10	**13.59**	**7.14**	-10.44	**2.70**	**2.23**	-10.29

The Datasets. We evaluate GUARD on **TOFU** (Maini et al., 2024) and **MUSE** (Shi et al., 2024) benchmarks for LLM unlearning (Jia et al., 2024; Ji et al., 2024). TOFU comprises 4,000 Q-A pairs from 200 synthetic author profiles. The data is split into: a) *Forget/Retain Sets*, sampled to unlearn 1%, 5%, or 10% of the data (the forget01, forget05, and forget10 splits); b) *Generalization Sets* (the *Real Author Set* and the *Real World Set*) to evaluate downstream retention and generalization. MUSE provides two real-world corpora: a) *NEWS (BBC articles)* with 0.8M/1.6M token Forget/Retain Sets; b) *BOOKS* with Harry Potter books as a 1.1M token Forget Set, and FanWiki as 0.5M token Retain Set. Each corpus includes text and knowledge sets with Q-A pairs derived from the original content.

The Backbone LLM. Following the original setups of the benchmarks, we evaluate unlearning performance using *Phi-1.5B* (Li et al., 2023) and *LLaMA-2-7B* (Touvron et al., 2023b) on TOFU, *LLaMA-2-7B* on MUSE NEWS corpus, and *ICLM-7B* (Shi et al., 2023) on MUSE BOOKS corpus.

Baselines. We compare against four non-loss-reweighting baselines (*GA* (Thudi et al., 2022), Gradient Difference (*GD*) (Liu et al., 2022), KL Minimization (*KM*) and Preference Optimization (*PO*) (Maini et al., 2024); PO is omitted for MUSE which is text-only), four loss-reweighting methods (*NPO* (Zhang et al., 2024a), *SimNPO* (Fan et al., 2024), *FLAT* (Wang et al., 2024) with the best-performing TV variant, *SatImp* (Yang et al., 2025)), and *RMU* (Li et al., 2024b). We apply GUARD on top of each non-loss-reweighting baseline and on top of NPO (chosen for stability); RMU is excluded from GUARD integration, since its forget-side term matches the activations of a single layer to a fixed random target, a representation-level objective that lies outside the analysis of Section 4. Appendices A and D.1 describe the baselines, and Appendix E.1 lists their hyperparameters.

Evaluation Metrics. We focus on metrics designed for the TOFU and MUSE benchmarks.

Evaluations of the TOFU dataset: To measure forgetting, retention, and the trade-off between the two, we evaluate the Absolute Performance and Sacrifice Rate.

(a) *Absolute Performance.* Given a dataset D and model θ, we use the Absolute Performance assessment $\epsilon(D;\theta)$ from the TOFU benchmark (ROUGE-L, Probability, Truth Ratio, all reported in percent). For ROUGE-L and Probability, a higher value on the Retain Set/lower value on the Forget Set indicate better retention/forgetting. The Truth Ratio (TR) measures the model's relative probability mass on perturbed answers vs. the correct ones; following the standard TOFU convention (Maini et al., 2024), it is reported as $\max(0, 1-\text{TR})$ on the Retain Set and as $\min(\text{TR}, 1/\text{TR})$ on the Forget Set, so that a *higher* reported Retain TR and a *higher* reported Forget TR are preferred.

(b) *Sacrifice Rate.* This metric, introduced in Eq. (7), is considered jointly with the Absolute Performance metric in order to capture both absolute and relative improvements. We denote by ρ_r, ρ_p, and ρ_t the Sacrifice Rates computed from ROUGE-L, Probability, and Truth Ratio, respectively. Sacrifice Rates are also reported

Table 3: Impact of GUARD on Absolute Performance evaluated over the Retain and Forget Sets for LLaMA-2-7B on 10% of TOFU data points. Per-metric arrows indicate the preferred direction following TOFU (Maini et al., 2024); bold letters indicate the better value within each (baseline, baseline+GUARD) pair for GA, KM, PO, and GD, and the best value among the loss-reweighting methods, RMU, and NPO+GUARD. GUARD consistently improves ROUGE-L and Probability on the Retain Set, and also improves the Retain Truth Ratio for GA, KM, and PO.

Methods	**Retain Set**			**Forget Set**		
	ROUGE-L (↑)	Prob. (%) (↑)	T.Ratio (↑)	ROUGE-L (↓)	Prob. (%) (↓)	T.Ratio (↑)
w/o Unlearn	98.57	98.95	47.47	98.54	99.09	51.60
GA	0.18	0.06	8.42	**0.32**	**0.07**	**83.85**
GA + GUARD	**22.40**	**17.50**	**23.11**	2.10	1.50	80.50
KM	0.18	0.05	7.18	**0.20**	**0.05**	**85.55**
KM + GUARD	**19.20**	**14.80**	**20.10**	1.90	1.30	82.30
PO	78.20	93.85	45.85	5.25	85.20	54.30
PO + GUARD	**84.50**	**97.30**	**46.80**	**4.20**	**82.50**	**60.50**
GD	47.15	55.30	**48.95**	**0.42**	0.12	**81.55**
GD + GUARD	**60.30**	**67.40**	47.80	0.95	**0.10**	80.40
NPO	78.50	92.30	49.80	35.20	38.50	67.30
SimNPO	80.20	93.10	49.20	34.80	39.20	65.80
FLAT (TV)	76.40	91.50	**49.85**	38.50	42.30	**70.20**
SatImp	77.30	91.80	49.30	33.80	38.10	65.10
RMU	84.50	95.20	49.10	**33.50**	**37.80**	64.50
NPO + GUARD	**89.30**	**97.20**	48.30	33.80	38.20	65.20

in percent. For the Truth Ratio, a higher value on the Forget Set indicates forgetting, so the sign of the denominator in Eq. (7) is reversed for ρ_t.

Evaluations of the MUSE dataset: Following the original MUSE framework, we evaluate methods according to four key criteria: (a) *Forget Set VerbMem.* This criterion measures exact text reproduction from the Forget Set. Lower values indicate better unlearning. (b) *Forget Set KnowMem.* This metric assesses the ability of the model to answer questions about the Forget Set. Lower values indicate better unlearning. (c) *PrivLeak.* This metric quantifies membership inference vulnerability. Values within $[-5\%, 5\%]$ indicate successful unlearning, while values outside this range suggest over/under-unlearning. (d) *Retain Set KnowMem.* This criterion is used to evaluate knowledge retention on the Retain Set. Higher values of the score indicate better utility preservation.

Configuration and Hyperparameters. We follow the protocol outlined by Maini et al. (2024) for all TOFU experiments: An AdamW optimizer with learning rate 10^{-5}, weight decay 0.01, 5 unlearning epochs, an effective batch size of 32, and a linear warm-up during the first epoch. All reported TOFU numbers correspond to the checkpoint after the final (5-th) unlearning epoch, matching the convention used by the official TOFU leaderboard. For MUSE, we adopt the standard MUSE configuration (Shi et al., 2024) (AdamW, lr $= 10^{-5}$, per-device batch size 32, up to 10 unlearning epochs with early stopping when utility drops below the released MUSE `Retrain` reference, i.e., the model retrained without the Forget Set). When applying GUARD on top of a baseline, we keep the baseline-specific hyperparameters at their published values and introduce only the GUARD temperature τ, isolating the effect of attribution-driven reweighting. We make two explicit changes: for NPO on MUSE, to boost privacy, and for RMU, for which we use the configuration of the `open-unlearning` codebase[1] (Appendix E.1).

Results. We evaluate the effectiveness of GUARD by focusing on the LLaMA-2-7B backbone, and defer the results for Phi-1.5B to Appendix E.3, as both models exhibit similar behaviors. We report on experimental results for both TOFU (Sacrifice Rates in Table 2 and Absolute Performance metrics in Table 3) and MUSE (four criteria in Table 4). Due to space limitations, in the main text we focus on results for 10% unlearning on TOFU as this is the most challenging scenario, while results for 1% and 5% data unlearning are delegated to Appendix E.2. For the same reason, we only present results for MUSE NEWS and delegate the MUSE BOOKS dataset analyses to Appendix E.4.

Performance of GUARD on TOFU. As shown in Table 2, GUARD consistently lowers the Sacrifice Rate across all baselines and evaluation sets wherever a baseline degrades performance (positive values). Where a baseline improves an evaluation set (negative values), GUARD keeps the change closer to zero. GA+GUARD reduces ρ_t on the Retain Set by up to 30.39% over GA. Table 3 further shows that on the Retain Set GA+GUARD improves ROUGE-L from 0.18 to 22.40 (a 124× increase) and Probability from

[1] `https://github.com/locuslab/open-unlearning`

Table 4: Unlearning experiments with LLaMA-2-7B on the MUSE NEWS dataset, "w/o Unlearn" is the released MUSE NEWS target model.

Methods	Forget Set VerbMem (↓)	Forget Set KnowMem (↓)	PrivLeak (∈ [−5%, 5%])	Retain Set KnowMem (↑)
		NEWS		
w/o Unlearn	58.4	63.9	-99.8	55.2
GA	**0.0**	**0.0**	**5.6**	0.0
GA + GUARD	**0.0**	**0.0**	6.4	**8.7**
KM	**27.8**	50.1	**-95.8**	43.9
KM + GUARD	28.3	**49.5**	-97.2	**50.4**
GD	4.6	31.8	**102.4**	27.8
GD + GUARD	**2.7**	**26.1**	106.5	**44.9**
NPO	1.5	44.8	93.0	37.2
SimNPO	**1.4**	44.5	91.0	41.5
FLAT (TV)	1.7	**43.8**	**84.5**	33.0
SatImp	1.6	45.0	95.5	43.9
RMU	1.5	44.0	95.0	43.1
NPO + GUARD	1.6	45.2	98.2	**50.0**

0.06% to 17.50%, demonstrating substantial recovery from GA collapse while maintaining comparable Forget Set degradation. All baselines are run under the same TOFU protocol with the hyperparameters of their official releases (for RMU, the `open-unlearning` configuration), so Tables 2–3 are direct comparisons. Among all loss-reweighting methods, NPO+GUARD achieves the lowest Retain ρ_r and ρ_p (14.32 and 2.87, reductions of 33.80% and 53.10% over the strongest baseline RMU at 21.63 and 6.12).

Performance of GUARD on MUSE. Table 4 shows that GUARD consistently improves retention performance across all baselines while maintaining effective forgetting, with NPO + GUARD achieving the best overall performance on Retain Set KnowMem (50.0) compared to all competing loss reweighting methods. We note that GUARD exhibits somewhat higher PrivLeak scores, which reflects the fact that retention of relevant knowledge naturally makes the model more vulnerable to membership inference attacks. Moreover, this moderate privacy degradation (a 5.59% leakage increase from NPO at 93.0 to NPO + GUARD at 98.2) is substantially outweighed by the significant gain in knowledge retention (a 34.41% improvement in Retain Set KnowMem, from 37.2 to 50.0), with comparable Forget Set KnowMem (45.2 vs. 44.8 for NPO).

Subsampled and Synthetic Retain Sets. A practical concern is whether GUARD relies on full access to the original Retain Set in order to compute its attribution score $a_i^{\text{GUARD}} = [J_{\text{avg}}^{D_r}]^\top J_i(\theta_0)$. We performed two ablations on TOFU forget10 (LLaMA-2-7B): (i) *Limited Retain*: The gradient $J_{\text{avg}}^{D_r}$ is computed on a 20% randomly sampled subset of D_r; and (ii) *Synthetic Retain*: D_r is replaced *entirely* by a GPT-4 generated proxy $\tilde{D}_r$ that only shares the general format and topic area of D_r (for TOFU, the format is QA pairs and the topic is fictional authors), but is nearly content-disjoint from D_f. The synthesis prompt only uses this information which is typically maintained even after compliance-driven data deletion (see Appendix E.7). In both cases, the unlearning step still runs on the original full data and only the attribution score is computed on the surrogate. As shown in Table 5, in both settings we preserve most of the full-retain improvements of GUARD over NPO: With a 20% retain sampling, NPO+GUARD reduces ρ_r and ρ_p from 31.69 and 10.97 (NPO) to 17.78 and 5.69 (vs. 14.32 and 2.87 at full-retain); with the synthetic proxy and no original retain data used in the attribution, ρ_r and ρ_p drop to 19.60 and 6.85, close to the result for 20%-subsampling and well below NPO. Retain ROUGE-L and Probability follow the same trend, while Forget Set metrics remain comparable across the four settings, confirming that surrogate-based attribution does not compromise forgetting. Experiments on MUSE NEWS and MUSE BOOKS are deferred to Appendix E.6 (Table 14).

Alternative Attribution Methods. We also compare the gradient-product attribution method of GUARD against three alternatives, namely feature-space Embedding Similarity (ES) and two first-order influence-function approximations, RelatIF (Barshan et al., 2020) and RIF (Rubinstein & Hopkins, 2025), all integrated into NPO under the same protocol. GUARD consistently achieves the lowest Retain ρ_r (14.32 vs. RIF's 19.08 and ES's 26.53). This result is intuitively justified: The unnormalized gradient dot product preserves both *directional alignment* and *magnitude information,* while, by Lemma 1 (Appendix B.2), dropping the inverse Hessian only rescales the score globally, which is absorbed by the temperature, up to a bounded residual. Experimental per-metric comparisons are reported in Appendix E.9.

Additional Results in the Appendix. We also provide a runtime analysis that shows that the

Table 5: GUARD used with subsampled (random, 20%) and synthetic (GPT-4 generated, *no* original retain data used in the attribution) Retain Sets on TOFU forget10 (LLaMA-2-7B). In both cases, GUARD retains its gains over NPO and, in Retain ROUGE-L and ρ_r, over RMU (which inherently requires knowledge of the Retain Set): even the synthetic-retain GUARD variant, using *no* original retain data in the attribution, outperforms RMU at full retain access in Retain ROUGE-L and ρ_r. The synthetic proxy is itself competitive with 20% random subsampling.

Methods	**Retain Set**					**Forget Set**		
	ROUGE-L (↑)	Prob. (%) (↑)	T. Ratio (↑)	ρ_r (↓)	ρ_p (↓)	ROUGE-L (↓)	Prob. (%) (↓)	T. Ratio (↑)
NPO	78.50	92.30	**49.80**	31.69	10.97	35.20	38.50	**67.30**
RMU (full retain)	84.50	95.20	49.10	21.63	6.12	**33.50**	**37.80**	64.50
NPO + GUARD (Full Retain)	**89.30**	**97.20**	48.30	**14.32**	**2.87**	33.80	38.20	65.20
NPO + GUARD (20% Retain)	87.20	95.50	48.65	17.78	5.69	34.59	38.47	64.81
NPO + GUARD (Synthetic Retain)	86.10	94.80	48.85	19.60	6.85	34.92	38.55	64.55

overhead of GUARD is small and incurred once only (Appendix E.5) and empirical visualizations of the proxy attribution scores and reweighting (Appendix E.8).

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Appendix

A Related Works

A.1 Unlearning LLMs

Existing LLM Unlearning approaches broadly fall into two categories: Fine-tuning-based and inference-time-based methods. Fine-tuning approaches such as Gradient Ascent (GA) (Thudi et al., 2022; Feng et al., 2024), Gradient Difference (GD) (Liu et al., 2022), KL Minimization (KM) and Preference Optimization (Maini et al., 2024), and variants thereof, modify model weights by adjusting training objectives, typically through loss maximization on the Forget Set (Zhang et al., 2024a) and incorporation of regularization terms based on KL divergence, retained data consistency, or prompt-guided responses (Xu et al., 2025; Bhaila et al., 2025). A complementary line of work explores inference-time unlearning, which avoids model updates by modifying prompts or logits at inference to suppress specific outputs (Liu et al., 2024a; Ji et al., 2024). While computationally attractive, these methods do not guarantee actual forgetting, as sensitive information remains embedded in the unchanged model parameters. Emerging directions (Yao et al., 2024b; Zhang et al., 2024b) also investigate properties beyond forgetting and retention, such as robustness of unlearning mechanisms and recovery from catastrophic forgetting.

Fine-Tuning-Based Methods. Many unlearning techniques modify model weights through loss-based fine-tuning strategies, such as GA (Feng et al., 2024), preference optimization (PO) (Maini et al., 2024), negative preference optimization (NPO) (Zhang et al., 2024a), and representation misdirection (Li et al., 2024b). RMU was introduced together with the WMDP benchmark for hazardous-knowledge removal, and it performs unlearning via representation-level (activation) misdirection while aiming to preserve benign capabilities (Li et al., 2024b). Methods of this type typically maximize the loss on the Forget Set while incorporating regularization objectives such as KL divergence minimization, joint training with retain data (Xu et al., 2025), or prompt-based desirable loss surrogates (Bhaila et al., 2025). Despite their effectiveness with respect to forgetting, these methods often significantly degrade performance on retained knowledge (Qiu et al., 2025). WAGLE (Jia et al., 2024) focuses on improving computation efficiency by isolating critical parameters for forgetting via weight attribution, whereas FLAT (Wang et al., 2024) adjusts the loss using only forget data, maximizing an f-divergence between template and forget responses, yet both empirically fail to preserve retention.

Loss-Reweighting Methods. Yang et al. (2025) analyze loss reweighting mechanisms through saturation and importance-based categorizations. NPO (Zhang et al., 2024a) treats Forget Data as negative responses in preference optimization, providing bounded objectives with adaptive gradient smoothing to prevent catastrophic collapse during unlearning. SimNPO (Fan et al., 2024) further addresses reference model bias in NPO by adopting a reference-free, length-normalized approach that more effectively allocates optimization power across forget samples with varying difficulty levels. Utility-guided methods, such as gradient-based influence modeling (Wang et al., 2025b), propose retention-aware metrics and derive from them the weighted GA (WGA) and token-wise NPO (TNPO) unlearning objectives.

Input-Based and In-Context Methods. Input-level methods operate at inference time by manipulating inputs to suppress undesired outputs without altering model weights. These include prompt filtering (Liu et al., 2024a) and logit-based interventions (Ji et al., 2024). ECO Prompts (Liu et al., 2024a) use prompt classifiers and zeroth-order optimization to induce forgetting, while Unlearning from Logit Difference (ULD) (Ji et al., 2024) employs an assistant LLM to reduce side effects. Although computationally efficient, these methods rely on access to sensitive data at inference time, which limits their privacy guarantees (Geng et al., 2025).

Alternative Directions. Recent efforts explore complementary directions beyond weight-based or input-level interventions. Multimodal unlearning (Li et al., 2024a), adversarial robustness (Schwinn et al., 2024), and standardized benchmarks (Wang et al., 2025a) enhance evaluation protocols but do not directly address

retention. Liu et al. (2025) survey the design space of LLM unlearning. Other approaches like quantization-based recovery (Zhang et al., 2024b) attempt to restore forgotten knowledge post hoc but remain orthogonal to the goal of preserving retention during unlearning.

A.2 Data Attribution

Data attribution in machine learning aims to quantify the influence of individual data points on model behavior, supporting tasks such as debugging, data cleaning, and interpretability. Classical approaches, such as influence functions, estimate the effect of removing a single data point on model parameters or predictions. Koh & Liang (2017) adapted influence functions to deep learning, illustrating their value in identifying mislabeled data and interpreting model decisions. However, subsequent studies revealed notable limitations. For example, Basu et al. (2020) showed that influence estimates are highly sensitive to hyperparameters and model architecture, particularly in deep networks. Hanawa et al. (2021) further found that these methods can fail basic "sanity checks," attributing high influence to unrelated data. Bae et al. (2022) also demonstrated that influence functions may estimate unintended parameters in deep neural networks, raising concerns about their reliability in complex models.

In response to these reliability issues, several variants have been proposed to refine the analytical influence formulation. To mitigate the tendency of influence functions to select outliers or mislabeled examples, Barshan et al. (2020) introduced RelatIF (Relative Influence). By normalizing the influence score against the self-influence of a training sample, RelatIF disentangles global impact from local relevance, prioritizing samples that are directionally aligned with the target prediction. More recently, addressing the approximation errors in high-dimensional settings, Rubinstein & Hopkins (2025) proposed Rescaled Influence Functions (RIF). RIF corrects the standard influence estimate using leverage scores, offering a more accurate prediction of the model change by explicitly accounting for the impact of data removal on the Hessian curvature.

To further address the limitations of classical influence functions, recent work has introduced more robust and scalable data attribution techniques. Among these, direct estimators such as regression-based data models have demonstrated improved predictive accuracy and generalization, particularly in large-scale settings where computational resources are sufficient (Ilyas et al., 2022). Two state-of-the-art approaches—TRAK (Park et al., 2023) and EK-FAC (Grosse et al., 2023)—offer practical solutions across diverse modalities. TRAK approximates a deep neural network with a linear surrogate model and estimates influence by ensembling over multiple model checkpoints and random projections. EK-FAC improves efficiency through a combination of strategies: partitioning the Hessian into block-diagonal components, expressing gradients as Kronecker products of smaller factors, and applying spectral regularization to stabilize curvature estimates.

Despite their empirical success, these methods remain computationally demanding for large-scale models such as large language models (LLMs). Moreover, their underlying attribution philosophy is not directly aligned with the goals of machine unlearning, which demands targeted forgetting and simultaneous preservation of nontargeted knowledge.

B Theoretical Guarantees of the Proposed Attribution-Tailored LLM Unlearning Approach

B.1 Preliminaries: Conventional Data Attribution

The Leave-One-Out Framework. Data attribution quantifies the impact of a specific training data point on the loss of the model under the Leave-One-Out framework (Rad & Maleki, 2018). Given a dataset D, the influence of a data point (x_j, y_j) on the optimal loss is defined as:

$$\Delta_L^j = \min_\theta L_D(\theta) - \min_\theta L_{D\setminus\{(x_j,y_j)\}}(\theta).$$

Influence Function. The *Influence Function* (Koh & Liang, 2017) provides an efficient approximation for the leave-one-out influence. Let $H_D(\theta)$ denote the Hessian matrix summing over data-wise second derivatives:

$H_D(\theta) := \sum_{(x_i,y_i)\in D} \nabla^2 l(x_i, y_i; \theta)$, The influence function for a data point (x_j, y_j) is then given by:

$$a_j^{\mathrm{IF}} = J_{avg}(\theta_0)^\top \cdot [H_D(\theta_0)]^{-1} \cdot J_j(\theta_0), \tag{13}$$

where $J_j(\theta) := \nabla_\theta l(x_j, y_j; \theta)$ represents the Jacobian of the loss function with respect to the model parameters, and $J_{avg}(\theta) := \frac{1}{|D|} \sum_{(x_i,y_i)\in D} J_i(\theta)$ is its average over D.

B.2 Influence Function Interpretation of the Proxy Score

Two choices in the design of GUARD deserve justification: dropping the inverse Hessian from the attribution score in Eq. (2), and down-weighting forget samples according to their alignment with the Retain Set in Eq. (3). This subsection addresses the first choice, and Section 4 addresses the second.

Compared to the influence function in Eq. (13), the proxy attribution score in Eq. (2) differs in two respects. First, the average gradient is taken over the Retain Set D_r rather than over the whole fine-tuning set D_0 (i.e., $D = D_0$ in Eq. (13)). This change is intended, since unlearning requires preserving the performance on D_r rather than on D_0. Second, the inverse Hessian is dropped. To quantify the effect of the second change, we compare a_j^{GUARD} with the influence function whose average is also taken over D_r, which we refer to as the *retention influence*:

$$a_j^{\mathrm{IF},r} := J_{\mathrm{avg}}^{D_r}(\theta_0)^\top H^{-1} J_j(\theta_0), \quad \forall j \in I_f, \qquad \text{where } H := H_{D_0}(\theta_0). \tag{14}$$

The retention influence approximates the effect of removing forget sample j on the average loss over the Retain Set, in the same leave-one-out sense as Eq. (13). Our analysis relies on the following assumption.

Assumption 3. *The loss function $l(x, y; \theta)$ is twice differentiable with respect to θ, and the Hessian matrix H is symmetric positive definite.*

Remark: When H is indefinite, as is common at LLM checkpoints, the damped Hessian $H + \mu I$ with μ large enough that $H + \mu I$ is positive definite (Koh & Liang, 2017) is used in place of H, and the results below apply to it.

Let $0 < \lambda_{\min} \le \lambda_{\max}$ denote the smallest and largest eigenvalues of H^{-1}, and let $\{v_k\}_k$ be an orthonormal eigenbasis of H^{-1}.

Lemma 1. *Suppose that Assumption 3 holds. Let $c := (\lambda_{\max} + \lambda_{\min})/2$ and $r_j := a_j^{IF,r} - c \cdot a_j^{GUARD}$. Then, for all $j \in I_f$,*

$$|r_j| \le \frac{\lambda_{\max} - \lambda_{\min}}{2} \cdot \left\|J_{avg}^{D_r}(\theta_0)\right\|_2 \cdot \left\|J_j(\theta_0)\right\|_2. \tag{15}$$

If, in addition, $(v_k^\top J_{avg}^{D_r}(\theta_0)) \cdot (v_k^\top J_j(\theta_0)) \ge 0$ for all k, then $\lambda_{\min} \cdot a_j^{GUARD} \le a_j^{IF,r} \le \lambda_{\max} \cdot a_j^{GUARD}$.

The proof is provided below.

Write $\bar{J} := J_{\mathrm{avg}}^{D_r}(\theta_0)$ and $J_j := J_j(\theta_0)$. Let $\{\lambda_k\}_k$ be the eigenvalues of H^{-1} associated with the orthonormal eigenvectors $\{v_k\}_k$, so that $\lambda_k \in [\lambda_{\min}, \lambda_{\max}]$ with $\lambda_{\min} > 0$. We define the eigenbasis components

$$q_k := (v_k^\top \bar{J}) \cdot (v_k^\top J_j), \qquad \text{so that} \qquad \sum_k q_k = \bar{J}^\top J_j = a_j^{\mathrm{GUARD}}, \qquad \sum_k \lambda_k q_k = \bar{J}^\top H^{-1} J_j = a_j^{\mathrm{IF},r}.$$

Proof. We first prove the bound on the residual. With $c = (\lambda_{\max} + \lambda_{\min})/2$, we have

$$r_j = a_j^{\mathrm{IF},r} - c \cdot a_j^{\mathrm{GUARD}} = \bar{J}^\top \big(H^{-1} - cI\big) J_j.$$

The matrix $H^{-1} - cI$ is symmetric with eigenvalues $\lambda_k - c \in [-(\lambda_{\max} - \lambda_{\min})/2,\ (\lambda_{\max} - \lambda_{\min})/2]$. Therefore, its operator norm is at most $(\lambda_{\max} - \lambda_{\min})/2$, and the Cauchy-Schwarz inequality gives

$$|r_j| \le \|H^{-1} - cI\|_{\mathrm{op}} \cdot \|\bar{J}\|_2 \cdot \|J_j\|_2 \le \frac{\lambda_{\max} - \lambda_{\min}}{2} \cdot \|\bar{J}\|_2 \cdot \|J_j\|_2.$$

For the second statement, if $q_k \ge 0$ for all k, then $\lambda_{\min} q_k \le \lambda_k q_k \le \lambda_{\max} q_k$ for every k. Summing over k gives $\lambda_{\min} \cdot a_j^{\mathrm{GUARD}} \le a_j^{\mathrm{IF},r} \le \lambda_{\max} \cdot a_j^{\mathrm{GUARD}}$. □

Lemma 1 shows that the retention influence equals a global rescaling $c \cdot a_j^{\mathrm{GUARD}}$ of the proxy score, with the same constant c for all forget samples, plus a residual r_j. Relative to $c \cdot \|J_{\mathrm{avg}}^{D_r}(\theta_0)\|_2 \|J_j(\theta_0)\|_2$, the size of the residual is at most $(\lambda_{\max} - \lambda_{\min})/(\lambda_{\max} + \lambda_{\min})$, which depends only on the condition number of H. The global rescaling does not affect GUARD: using $c \cdot a_j^{\mathrm{GUARD}}$ in Eq. (3) with temperature τ' gives the same weights as using a_j^{GUARD} with temperature $\tau = \tau'/c$. Hence the constant c is absorbed into the temperature and never needs to be estimated, and only the residual changes the weights, each by a factor in $[e^{-2R/\tau'}, e^{2R/\tau'}]$ with $R := \max_{j \in I_f} |r_j|$ (Appendix B.3). Dropping the inverse Hessian is therefore exact when H^{-1} acts as a multiple of the identity on the span of the gradients, and otherwise leads to a bounded perturbation of the weights.

B.3 Effect of the Residual on the Unlearning Weights

Let $\omega_i^{\mathrm{IF}} := n_f \cdot e^{-a_i^{\mathrm{IF},r}/\tau'} / \sum_{k \in I_f} e^{-a_k^{\mathrm{IF},r}/\tau'}$ be the weights obtained from the retention influence at temperature τ', and let $\omega_i := n_f \cdot e^{-c\, a_i^{\mathrm{GUARD}}/\tau'} / \sum_{k \in I_f} e^{-c\, a_k^{\mathrm{GUARD}}/\tau'}$ be the GUARD weights at temperature $\tau = \tau'/c$. With $R := \max_{k \in I_f} |r_k|$, we have

$$\frac{\omega_i^{\mathrm{IF}}}{\omega_i} = e^{-r_i/\tau'} \cdot \frac{\sum_k e^{-c\, a_k^{\mathrm{GUARD}}/\tau'}}{\sum_k e^{-c\, a_k^{\mathrm{GUARD}}/\tau'}\, e^{-r_k/\tau'}} \in \left[e^{-2R/\tau'},\, e^{2R/\tau'}\right],$$

since the first factor lies in $[e^{-R/\tau'}, e^{R/\tau'}]$ and the second factor is the reciprocal of a weighted average of the terms $e^{-r_k/\tau'}$, which lie in the same interval. Hence, the global constant c is absorbed into the temperature, and only the residual changes the weights, multiplicatively and by a bounded factor. GUARD does not need to estimate c, since τ is a hyperparameter.

C Proof of Results in Section 4

C.1 Proof of Theorem 1

Recall that the losses of different models on D_r are defined as

$$L_{D_r}^0 = \frac{1}{n_r} \sum_{i \in I_r} l(x_i, y_i; \theta^0), \quad L_{D_r}^{\mathrm{GA}} = \frac{1}{n_r} \sum_{i \in I_r} l(x_i, y_i; \theta^{GA}), \quad L_{D_r}^{\mathrm{GUARD}} = \frac{1}{n_r} \sum_{i \in I_r} l(x_i, y_i; \theta^{\mathrm{GUARD}}).$$

The model update rules given by GA and by GUARD are,

$$\theta^{\mathrm{GA}} = \theta^0 + \eta \cdot \frac{1}{n_f} \sum_{i \in I_f} g_i, \quad \theta^{\mathrm{GUARD}} = \theta^0 + \eta \cdot \frac{1}{n_f} \sum_{i \in I_f} [\omega_i^{\mathrm{GUARD}} \cdot g_i],$$

where $\omega_i^{\mathrm{GUARD}}$ for $i \in I_f$ is defined in (3). Also recall that $a_j = \langle \bar{g}_r, g_j \rangle$, and thus we have

$$\frac{1}{n_f} \sum_{j \in I_f} a_j = \kappa, \quad \frac{1}{n_f} \sum_{j \in I_f} a_j^2 = \sigma_\kappa^2 + \kappa^2. \tag{16}$$

From the loss definitions and model update rules, we get the approximations of the increase in the retention loss compared with the model before unlearning. Specifically, both updates have norm at most ηG, since the GUARD weights are nonnegative and average to one. Hence, by Assumption 2, using a first-order Taylor expansion around θ_0, the increase in the loss on the D_r induced by GA is given by the first-order approximation

$$\begin{aligned} L_{D_r}^{\mathrm{GA}} - L_{D_r}^0 &= \frac{1}{n_r} \sum_{i \in I_r} g_i^\top \cdot [\theta^{\mathrm{GA}} - \theta^0] + O(\eta^2 G^2) \\ &= \frac{1}{n_r} \sum_{i \in I_r} g_i^\top \cdot [\eta \cdot \frac{1}{n_f} \sum_{j \in I_f} g_j] + O(\eta^2 G^2) \\ &= \eta \cdot \langle \bar{g}_f, \bar{g}_r \rangle + O(\eta^2 G^2) \\ &= \eta \cdot \kappa + O(\eta^2 G^2). \end{aligned} \tag{17}$$

Similarly, the increase in the loss on D_r induced by GUARD is given by

$$\begin{aligned} L_{D_r}^{\text{GUARD}} - L_{D_r}^0 &= \frac{1}{n_r} \sum_{i \in I_r} g_i^\top \cdot [\theta^{\text{GUARD}} - \theta^0] + O(\eta^2 G^2) \\ &= \frac{1}{n_r} \sum_{i \in I_r} g_i^\top \cdot [\eta \cdot \frac{1}{n_f} \sum_{j \in I_f} [\omega_j^{\text{GUARD}} \cdot g_j]] + O(\eta^2 G^2). \end{aligned} \tag{18}$$

Writing $\overline{g}_f' := \frac{1}{n_f} \sum_{i \in I_f} \left[\omega_i^{\text{GUARD}} \cdot g_i\right]$, we can express the above equation as

$$L_{D_r}^{\text{GUARD}} - L_{D_r}^0 = \eta \cdot \langle \overline{g}_r, \overline{g}_f' \rangle + O(\eta^2 G^2). \tag{19}$$

Using the Taylor expansion $e^{-x} = 1 - x + O(x^2)$, ω_i^{GUARD} becomes

$$\begin{aligned} \omega_i^{\text{GUARD}} =& n_f \frac{e^{-a_i/\tau}}{\sum_{k \in I_f} e^{-a_k/\tau}} \\ =& n_f \frac{1 - a_i/\tau + O(\delta_a^2)}{\sum_{k \in I_f} (1 - a_k/\tau + O(\delta_a^2))} \\ =& \frac{1 - a_i/\tau + O(\delta_a^2)}{1 - \frac{\kappa}{\tau} + O(\delta_a^2)} \\ =& \frac{1 - a_i/\tau}{1 - \frac{\kappa}{\tau}} + O(\delta_a^2), \end{aligned} \tag{20}$$

where we use $|a_k|/\tau \leq \delta_a$ for all $k \in I_f$, and the last line holds since $1 - \kappa/\tau \geq 1 - \delta_a$ is bounded away from zero under Assumption 1, as κ is the average of the a_j. We denote the remainder in the last line by R_i, so that $|R_i| = O(\delta_a^2)$. Based on the above expression, and since $\langle \overline{g}_r, g_j \rangle = a_j$, we can simplify the first term in (19) as

$$\begin{aligned} \langle \overline{g}_r, \overline{g}_f' \rangle =& \frac{1}{1 - \kappa/\tau} \left[\frac{1}{n_f} \sum_{j \in I_f} a_j - \frac{1}{\tau} \frac{1}{n_f} \sum_{j \in I_f} a_j^2 \right] + O(\tau\, \delta_a^3) \\ \overset{(a)}{=}& \frac{\kappa - \frac{1}{\tau}(\sigma_\kappa^2 + \kappa^2)}{1 - \kappa/\tau} + O(\tau\, \delta_a^3) \\ =& \kappa - \frac{\sigma_\kappa^2}{\tau - \kappa} + O(\tau\, \delta_a^3), \end{aligned} \tag{21}$$

where (a) follows from (16), and the remainder equals $\frac{1}{n_f} \sum_{j \in I_f} R_j a_j = O(\tau\, \delta_a^3)$, since $|a_j| \leq \tau \delta_a$. Substituting (21) into (19), we have

$$L_{D_r}^{\text{GUARD}} - L_{D_r}^0 = \eta \left(\kappa - \frac{\sigma_\kappa^2}{\tau - \kappa} \right) + \eta \cdot O(\tau\, \delta_a^3) + O(\eta^2 G^2). \tag{22}$$

Hence, we can conclude that

$$L_{D_r}^{\text{GA}} - L_{D_r}^{\text{GUARD}} = (1 - \delta_\kappa)^{-1} \cdot \eta/\tau \cdot \sigma_\kappa^2 + \eta \cdot O(\tau\, \delta_a^3) + O(\eta^2 G^2).$$

C.2 Proof of Theorem 2

Recall that the losses of different models on the Forget Set are defined as

$$\begin{aligned} L_{D_f}^0 &= \frac{1}{n_f} \sum_{i \in I_f} l(x_i, y_i; \theta^0), \\ L_{D_f}^{\text{GA}} &= \frac{1}{n_f} \sum_{i \in I_f} l(x_i, y_i; \theta^{GA}), \\ L_{D_f}^{\text{GUARD}} &= \frac{1}{n_f} \sum_{i \in I_f} l(x_i, y_i; \theta^{\text{GUARD}}). \end{aligned}$$

Correspondingly, the model update rules given by GA and by GUARD are, respectively,

$$\theta^{\mathrm{GA}} = \theta^0 + \eta \cdot \frac{1}{n_f} \sum_{i \in I_f} g_i$$
$$\theta^{\mathrm{GUARD}} = \theta^0 + \eta \cdot \frac{1}{n_f} \sum_{i \in I_f} [\omega_i^{\mathrm{GUARD}} \cdot g_i].$$

As in the proof of Theorem 1, both updates have norm at most ηG, so by Assumption 2 the second-order Taylor remainders are $O(\eta^2 G^2)$. Hence, based on loss definitions and model update rules, we get the approximations of the increase in the forget loss compared with the model before unlearning. Specifically, the increase in the loss on the Forget Set induced by GA is given by

$$\begin{aligned} L_{D_f}^{\mathrm{GA}} - L_{D_f}^0 &= \frac{1}{n_f} \sum_{i \in I_f} g_i^\top \cdot [\theta^{\mathrm{GA}} - \theta^0] + O(\eta^2 G^2) \\ &= \frac{1}{n_f} \sum_{i \in I_f} g_i^\top \cdot [\eta \cdot \frac{1}{n_f} \sum_{j \in I_f} g_j] + O(\eta^2 G^2) \\ &= \eta \cdot \langle \overline{g}_f, \overline{g}_f \rangle + O(\eta^2 G^2), \end{aligned} \tag{23}$$

and the increase in the loss on the Forget Set induced by GUARD is given by

$$\begin{aligned} L_{D_f}^{\mathrm{GUARD}} - L_{D_f}^0 &= \frac{1}{n_f} \sum_{i \in I_f} g_i^\top \cdot [\theta^{\mathrm{GUARD}} - \theta^0] + O(\eta^2 G^2) \\ &= \frac{1}{n_f} \sum_{i \in I_f} g_i^\top \cdot \left[\eta \cdot \frac{1}{n_f} \sum_{j \in I_f} [\omega_j^{\mathrm{GUARD}} \cdot g_j] \right] + O(\eta^2 G^2) \\ &= \eta \cdot \langle \overline{g}'_f, \overline{g}_f \rangle + O(\eta^2 G^2), \end{aligned} \tag{24}$$

where $\overline{g}'_f$ is the weighted average gradient over Forget Set, $\overline{g}'_f := \frac{1}{n_f} \sum_{i \in I_f} [\omega_i^{\mathrm{GUARD}} \cdot g_i]$. Also,

$$\begin{aligned} \langle \overline{g}_f, \overline{g}'_f \rangle &= \left[\frac{1}{n_f} \sum_{i \in I_f} g_i \right]^\top \cdot \left[\frac{1}{n_f} \sum_{j \in I_f} [\omega_j^{\mathrm{GUARD}} \cdot g_j] \right] \\ &= \frac{1}{{n_f}^2} \sum_{i,j \in I_f} \left[n_f \cdot \frac{e^{-a_j^{\mathrm{GUARD}}/\tau}}{\sum_{k \in I_f} e^{-a_k^{\mathrm{GUARD}}/\tau}} \cdot g_i^\top \cdot g_j \right] \\ &= \frac{1}{n_f} \sum_{i,j \in I_f} \left[\frac{1 - \overline{g}_r^\top \cdot g_j/\tau}{\sum_{k \in I_f} [1 - \overline{g}_r^\top \cdot g_k/\tau]} \cdot g_i^\top \cdot g_j \right] + O(\delta_a^2 G^2) \\ &= \frac{1}{{n_f}^2} \cdot \frac{\tau}{\tau - \kappa} \sum_{i,j \in I_f} \left[[1 - \overline{g}_r^\top \cdot g_j/\tau] \cdot g_i^\top \cdot g_j \right] + O(\delta_a^2 G^2) \\ &= \frac{\tau}{n_f(\tau - \kappa)} \left[n_f \|\overline{g}_f\|_2^2 - \frac{1}{\tau} \sum_{j \in I_f} \left[\overline{g}_r^\top \cdot g_j \cdot g_j^\top \cdot \overline{g}_f \right] \right] + O(\delta_a^2 G^2). \end{aligned} \tag{25}$$

Here, the third line uses the expansion of $\omega_j^{\mathrm{GUARD}}$ from the proof of Theorem 1, and its remainder equals $\frac{1}{n_f} \sum_{j \in I_f} R_j \, g_j^\top \overline{g}_f = O(\delta_a^2 G^2)$, since $|g_j^\top \overline{g}_f| \le G^2$. Let d be the dimension of the gradient vectors. Then we

can write

$$\begin{aligned}\sum_{j\in I_f} \overline{g}_r^\top g_j \cdot g_j^\top \overline{g}_f &= \overline{g}_r^\top \left(\sum_{j\in I_f} g_j g_j^\top\right) \overline{g}_f \\ &= n_f \cdot \overline{g}_r^\top \Sigma_f \overline{g}_f \\ &= n_f \bar{g}_r^\top \left(\mathrm{Cov}_f + \bar{g}_f \bar{g}_f^\top\right) \bar{g}_f \\ &= n_f \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f + n_f \left\langle \bar{g}_r, \bar{g}_f \right\rangle \|\bar{g}_f\|_2^2 \\ &= n_f \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f + n_f \, \kappa \, \|\bar{g}_f\|_2^2.\end{aligned}$$

Thus, we have

$$\begin{aligned}\langle \overline{g}'_f, \overline{g}_f \rangle =& \frac{\tau}{\tau-\kappa}\|\bar{g}_f\|_2^2 - \frac{1}{\tau-\kappa}\, \bar{g}_r^\top \Sigma_f \bar{g}_f + O(\delta_a^2 G^2) \\ =& \frac{\tau}{\tau-\kappa}\|\bar{g}_f\|_2^2 - \frac{1}{\tau-\kappa}\left(\kappa \|\bar{g}_f\|_2^2 + \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f\right) + O(\delta_a^2 G^2) \\ =& \|\bar{g}_f\|_2^2 - \frac{1}{\tau-\kappa} \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f + O(\delta_a^2 G^2).\end{aligned} \tag{26}$$

Substituting Eq. (26) into Eq. (24), we get

$$L_{D_f}^{\mathrm{GUARD}} - L_{D_f}^0 = \eta \langle \bar{g}'_f, \bar{g}_f \rangle + O(\eta^2 G^2) = \eta \|\bar{g}_f\|_2^2 - \frac{\eta}{\tau-\kappa}\, \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f + \eta \cdot O(\delta_a^2 G^2) + O(\eta^2 G^2). \tag{27}$$

Hence, using Eq. (27) and Eq. (23), we have

$$L_{D_f}^{\mathrm{GA}} - L_{D_f}^{\mathrm{GUARD}} = \frac{\eta}{\tau-\kappa}\, \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f + \eta \cdot O(\delta_a^2 G^2) + O(\eta^2 G^2). \tag{28}$$

Under Assumption 1, we have $\tau > \kappa$ (since $\kappa/\tau \le \delta_a \ll 1$), and we have

$$\begin{aligned}\left| L_{D_f}^{\mathrm{GA}} - L_{D_f}^{\mathrm{GUARD}} \right| =& \frac{\eta}{\tau-\kappa} \left| \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f \right| + \eta \cdot O(\delta_a^2 G^2) + O(\eta^2 G^2) \\ \le& \frac{\eta}{\tau-\kappa} \sqrt{\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_r} \sqrt{\bar{g}_f^\top \mathrm{Cov}_f \bar{g}_f} + \eta \cdot O(\delta_a^2 G^2) + O(\eta^2 G^2),\end{aligned} \tag{29}$$

where the last line follows from the Cauchy-Schwarz inequality

$$\left| \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f \right| \le \sqrt{\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_r} \sqrt{\bar{g}_f^\top \mathrm{Cov}_f \bar{g}_f}. \tag{30}$$

By definition of Cov_f, we have

$$\begin{aligned}\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_r =& \frac{1}{n_f} \sum_{i\in I_f} \bar{g}_r^\top (g_i - \overline{g}_f)(g_i - \overline{g}_f)^\top \bar{g}_r \\ =& \frac{1}{n_f} \sum_{i\in I_f} \left(\bar{g}_r^\top (g_i - \overline{g}_f)\right)^2 \\ =& \frac{1}{n_f} \sum_{i\in I_f} (a_i - \kappa)^2 \\ =& \sigma_\kappa^2.\end{aligned} \tag{31}$$

Thus,

$$\left| \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f \right| \le \sqrt{\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_r} \sqrt{\bar{g}_f^\top \mathrm{Cov}_f \bar{g}_f} \le \sigma_\kappa \sqrt{\bar{g}_f^\top \mathrm{Cov}_f \bar{g}_f}. \tag{32}$$

We can further bound the RHS using the spectrum of Cov_f. Specifically, we have

$$\bar{g}_f^\top \mathrm{Cov}_f \bar{g}_f \leq \lambda_{\max}(\mathrm{Cov}_f)\, \|\bar{g}_f\|_2^2, \tag{33}$$

and also

$$\left|\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f\right| \leq \sigma_\kappa \sqrt{\lambda_{\max}(\mathrm{Cov}_f)} \|\bar{g}_f\|_2. \tag{34}$$

Hence, we conclude

$$\left| L_{D_f}^{\mathrm{GA}} - L_{D_f}^{\mathrm{GUARD}} \right| \leq \frac{\eta \sigma_\kappa}{\tau - \kappa} \sqrt{\lambda_{\max}(\mathrm{Cov}_f)} \|\bar{g}_f\|_2 + \eta \cdot O(\delta_a^2 G^2) + O(\eta^2 G^2). \tag{35}$$

C.3 Proof of Theorem 3

Throughout this proof, we work to leading order, i.e., we drop the remainder terms in Theorems 1 and 2, and ρ^{GA} and ρ^{GUARD} denote the resulting leading-order Sacrifice Rates. From Eq. (17) and Eq. (23), we have

$$\rho^{\mathrm{GA}} = \frac{L_{D_r}^{\mathrm{GA}} - L_{D_r}^0}{L_{D_f}^{\mathrm{GA}} - L_{D_f}^0} = \frac{\eta \cdot \kappa}{\eta \cdot \langle \bar{g}_f, \bar{g}_f \rangle} = \frac{\kappa}{\|\bar{g}_f\|_2^2}, \tag{36}$$

which is well defined since $\bar{g}_f \neq 0$ under condition (8).

To compute ρ^{GUARD}, using Eq. (22) and Eq. (27) we get

$$\begin{aligned} \rho^{\mathrm{GUARD}} &= \frac{L_{D_r}^{\mathrm{GUARD}} - L_{D_r}^0}{L_{D_f}^{\mathrm{GUARD}} - L_{D_f}^0} \\ &= \frac{\eta \left(\kappa - \frac{\sigma_\kappa^2}{\tau - \kappa} \right)}{\eta \left(\|\bar{g}_f\|_2^2 - \frac{1}{\tau - \kappa} \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f \right)} \\ &= \frac{\kappa - \frac{\sigma_\kappa^2}{\tau - \kappa}}{\|\bar{g}_f\|_2^2 - \frac{1}{\tau - \kappa} \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f}, \end{aligned} \tag{37}$$

which is well defined if its denominator is positive. Using (34), a sufficient condition for this to hold true is

$$\|\bar{g}_f\|_2^2 > \frac{\sigma_\kappa}{\tau - \kappa} \sqrt{\lambda_{\max}(\mathrm{Cov}_f)} \|\bar{g}_f\|_2. \tag{38}$$

That is, if $\|\bar{g}_f\|_2 > \frac{\sigma_\kappa}{\tau - \kappa} \sqrt{\lambda_{\max}(\mathrm{Cov}_f)}$, then we have

$$\rho^{\mathrm{GA}} - \rho^{\mathrm{GUARD}} = \frac{\|\bar{g}_f\|_2^2 \sigma_\kappa^2 - \kappa\, \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f}{(\tau - \kappa)\, \|\bar{g}_f\|_2^2 \left(\|\bar{g}_f\|_2^2 - \frac{1}{\tau - \kappa}\, \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f \right)}. \tag{39}$$

From the above expression, GUARD is better than GA (to leading order), when $\rho^{\mathrm{GA}} - \rho^{\mathrm{GUARD}} \geq 0$. We next discuss two cases. First, when $\sigma_\kappa = 0$, $a_j = \kappa$ for all $j \in I_f$, i.e., every forget gradient has the same alignment with the retain gradient. In this case, the GUARD weights become uniform, and the GUARD reduces to the GA updates. When $\sigma_\kappa \neq 0$, since we already ensured that the denominator is positive, this reduces to the numerator being nonnegative. Using (34), a sufficient condition is

$$\|\bar{g}_f\|_2 \geq \frac{\kappa}{\sigma_\kappa} \sqrt{\lambda_{\max}(\mathrm{Cov}_f)}. \tag{40}$$

The denominator is positive if $\|\bar{g}_f\|_2 > \frac{\sigma_\kappa}{\tau - \kappa} \sqrt{\lambda_{\max}(\mathrm{Cov}_f)}$. Putting these together, a sufficient condition is given by

$$\|\bar{g}_f\|_2 > \max\left\{ \frac{\sigma_\kappa}{\tau - \kappa}, \frac{\kappa}{\sigma_\kappa} \right\} \sqrt{\lambda_{\max}(\mathrm{Cov}_f)}. \tag{41}$$

This is condition (8) in Theorem 3.

Under the spiked covariance model with $\mathrm{Cov}_f = \nu^2 I + U\Lambda U^\top$, $\lambda_{\max}(\mathrm{Cov}_f) = \nu^2 + \lambda_{\max}(\Lambda)$. We hence obtain a sufficient condition for $\rho^{\mathrm{GA}} - \rho^{\mathrm{GUARD}} \geq 0$ as

$$\|\bar{g}_f\|_2 > \max\left\{\frac{\sigma_\kappa}{\tau-\kappa}, \frac{\kappa}{\sigma_\kappa}\right\}\sqrt{\nu^2 + \lambda_{\max}(\Lambda)}. \tag{42}$$

To obtain a sharper bound, note that

$$\bar{g}_f^\top \mathrm{Cov}_f \bar{g}_f = \nu^2\|\bar{g}_f\|_2^2 + \bar{g}_f^\top U\Lambda U^\top \bar{g}_f \leq \nu^2\|\bar{g}_f\|_2^2 + \lambda_{\max}(\Lambda)\|U^\top \bar{g}_f\|_2^2. \tag{43}$$

Hence,

$$\left|\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f\right| \leq \sigma_\kappa\sqrt{\nu^2\|\bar{g}_f\|_2^2 + \lambda_{\max}(\Lambda)\|U^\top \bar{g}_f\|_2^2}. \tag{44}$$

Thus, by the argument for the numerator above, the second condition in (8) can be relaxed to

$$\|\bar{g}_f\|_2^2\sigma_\kappa \geq \kappa\sqrt{\nu^2\|\bar{g}_f\|_2^2 + \lambda_{\max}(\Lambda)\|U^\top \bar{g}_f\|_2^2}. \tag{45}$$

Dividing both sides by $\|\bar{g}_f\|_2$ and writing $\cos^2\phi := \|U^\top \bar{g}_f\|_2^2/\|\bar{g}_f\|_2^2$ shows that this is condition (9).

The Special Case of Isotropic Gradients. Let $\mathrm{Cov}_f = \nu^2 I$ for some $\nu > 0$, i.e., $\Lambda = 0$ in the spiked model. By (31), $\sigma_\kappa^2 = \bar{g}_r^\top \mathrm{Cov}_f \bar{g}_r = \nu^2\|\bar{g}_r\|_2^2$, which is positive since $\kappa > 0$ implies $\bar{g}_r \neq 0$. Hence, by Theorem 1, to first order GUARD reduces the loss on the Retain Set by $\eta\nu^2\|\bar{g}_r\|_2^2/(\tau-\kappa)$ compared to GA. Moreover, $\bar{g}_r^\top \mathrm{Cov}_f \bar{g}_f = \nu^2\kappa$, so the numerator of (39) equals $\nu^2\big(\|\bar{g}_f\|_2^2\|\bar{g}_r\|_2^2 - \kappa^2\big)$, which is nonnegative by the Cauchy-Schwarz inequality in $\mathbb{R}^d$. Thus the second condition in (8) is not needed. The denominator of (39) equals $(\tau-\kappa)\|\bar{g}_f\|_2^2\big(\|\bar{g}_f\|_2^2 - \nu^2\kappa/(\tau-\kappa)\big)$, which is positive if and only if $\|\bar{g}_f\|_2^2 > \nu^2\kappa/(\tau-\kappa)$. Hence, under this condition, the leading-order term of $\rho^{\mathrm{GA}} - \rho^{\mathrm{GUARD}}$ is nonnegative.

C.4 Empirical Verification of the Sufficient Condition on TOFU

The sacrifice rate guarantee of Theorem 3 relies on the sufficient condition

$$\|\bar{g}_f\|_2 \;>\; \max\left\{\frac{\sigma_\kappa}{\tau-\kappa},\; \frac{\kappa}{\sigma_\kappa}\right\}\sqrt{\lambda_{\max}(\mathrm{Cov}_f)}, \tag{46}$$

together with Assumption 1 (positive global alignment $\kappa > 0$ and weak per-sample alignment $|a_j|/\tau \ll 1$). To check if this condition is realistic in the LLM domains that GUARD operates, we estimate every quantity in (46) using the full TOFU forget10 split and random subsamples of the retain split on the LLaMA-2-7B checkpoint fine-tuned under the official TOFU protocol, using the same temperature $\tau = 0.03$ adopted throughout the paper.

Estimation protocol. For all $n_f = 400$ forget points of the TOFU forget10 split and a random subsample of $n_r = 400$ retain points:

1. Compute per-sample gradients $\{g_i\}_{i\in I_f}, \{g_j\}_{j\in I_r}$ at the fine-tuned checkpoint θ_0.
2. Form the means $\bar{g}_f = \frac{1}{n_f}\sum_{i\in I_f} g_i$, $\bar{g}_r = \frac{1}{n_r}\sum_{j\in I_r} g_j$.
3. Compute scalars $\kappa = \langle \bar{g}_f, \bar{g}_r\rangle$, $\|\bar{g}_f\|_2$, and per-sample alignments $a_j = \langle \bar{g}_r, g_j\rangle$ for $j \in I_f$.
4. Compute the alignment variance $\sigma_\kappa^2 = \frac{1}{n_f}\sum_j a_j^2 - \kappa^2$.
5. Form the centered forget-gradient matrix $\tilde{G} \in \mathbb{R}^{n_f\times d}$ with rows $g_i - \bar{g}_f$, and obtain $\lambda_{\max}(\mathrm{Cov}_f)$ as the largest eigenvalue of the $n_f \times n_f$ Gram matrix $\frac{1}{n_f}\tilde{G}\tilde{G}^\top$ (the non-zero spectrum of $\mathrm{Cov}_f = \frac{1}{n_f}\tilde{G}^\top\tilde{G}$ is identical to that of $\frac{1}{n_f}\tilde{G}\tilde{G}^\top$, so the eigendecomposition is performed over $\mathbb{R}^{n_f\times n_f}$ rather than $\mathbb{R}^{d\times d}$).

The check on MUSE BOOKS in Figure 2 follows the same estimation steps.

Empirical evaluation over 50 random subsamples. We repeat the protocol above for 50 independent random retain subsamples of size $n_r = 400$, using all $n_f = 400$ forget points in each trial. For each subsample we compute every quantity in (46) and record the gap LHS$-$RHS $= \|\bar{g}_f\|_2 - \max\{\sigma_\kappa/(\tau-\kappa),\ \kappa/\sigma_\kappa\}\sqrt{\lambda_{\max}(\mathrm{Cov}_f)}$; a positive gap indicates that the sufficient condition is satisfied for that subsample. The empirical distribution of the gap across the 50 trials is shown in Figure 3.

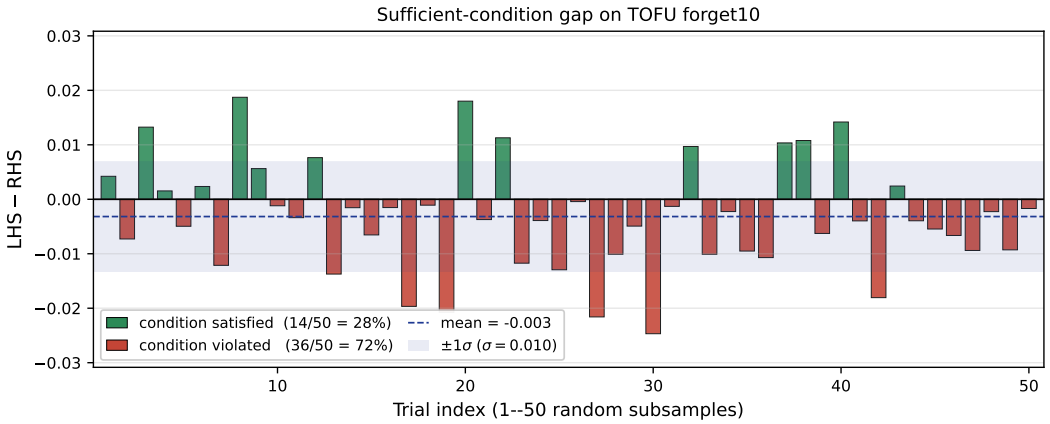


Figure 3: Empirical distribution of LHS $-$ RHS in the sufficient condition (46) across 50 trials, each using the full TOFU forget10 split ($n_f = 400$) and a random retain subsample ($n_r = 400$) (LLaMA-2-7B, $\tau = 0.03$). Green bars indicate trials where the sufficient condition is satisfied, red bars indicate violations.

While Theorem 3 provides a conservative sufficient condition, Figure 3 shows that condition (46) is indeed satisfied on a subset of subsamples (28%), with violations being small ($|\text{LHS} - \text{RHS}| < 0.025$). Moreover, even when the sufficient condition is violated, GUARD still demonstrates improved empirical performance, improving over GA in every Sacrifice Rate at the 10% split, as shown in Table 2.

D Algorithms

D.1 Update Rules of Baseline Methods

We list the baseline unlearning algorithms included in the TOFU benchmark (Maini et al., 2024): Gradient Ascent (GA), Gradient Difference (GD), KL Minimization (KM), and Preference Optimization (PO). We also list Negative Preference Optimization (NPO), the loss-reweighting method that we combine with GUARD.

Gradient Ascent (GA). The GA method aims to reduce the confidence of the model in its initial predictions on the Forget Set. Specifically, for each instance in D_f, the goal is to maximize the training loss, encouraging the model to diverge from its prior learned outputs, and the overall loss objective is defined as:

$$L_{GA} = -L_{D_f}(\theta) \tag{47}$$

Gradient Difference (GD) (Liu et al., 2022). The Gradient Difference method builds upon the gradient ascent approach by not only increasing loss on the Forget Set D_f but also preserving model performance on the Retain set D_r. The objective function for this approach is formulated as:

$$L_{\text{GD}} = -L_{D_f}(\theta) + L_{D_r}(\theta). \tag{48}$$

To balance computational cost, for each sample processed from D_f, a corresponding example from D_r is selected, ensuring the model remains within feasible resource constraints.

KL Divergence Minimization (KM). The KL Minimization technique aims to regulate the model's unlearning process by ensuring that predictions on the Retain set D_r remain consistent with those of the original model while still reducing accuracy on D_f. Let M be the model function, where $M(\cdot)$ produces a

probability distribution over possible token predictions. The training objective is expressed as:

$$L_{\mathrm{KM}} = -L_{D_f}(\theta) + \frac{1}{|D_r|}\sum_{s\in D_r}\frac{1}{|s|}\sum_{i=2}^{|s|}\mathrm{KL}\big(M_0(s_{<i}) \parallel M_{\mathrm{u}}(s_{<i})\big). \tag{49}$$

Here, M_0 and M_{u} refer to the model before and after unlearning, respectively. Due to resource constraints, instances from D_r are sampled randomly, while the entire Forget Set is used.

Preference Optimization (PO). Inspired by Direct Preference Optimization (DPO) (Rafailov et al., 2023), this method substitutes responses in D_f with alternative neutral responses such as "I do not have that information". The loss function is as follows:

$$L_{\mathrm{PO}} = L_{D_r}(\theta) + L_{D_f^{\mathrm{idk}}}(\theta). \tag{50}$$

This ensures that while the model updates its responses for D_f, its overall linguistic abilities and predictive accuracy on D_r remain stable.

Negative Preference Optimization (NPO) (Zhang et al., 2024a). NPO treats the forget samples as negative responses in a preference objective, which bounds the per-sample unlearning signal. We use NPO with a retain loss, i.e., the retain-regularized variant of (Zhang et al., 2024a), which corresponds to `NPO_GDR` in the MUSE codebase:

$$L_{\mathrm{NPO}} = \frac{1}{n_f}\sum_{i\in I_f}\frac{2}{\beta}\log\left(1+\left(\frac{\pi_\theta(y_i|x_i)}{\pi_{\mathrm{ref}}(y_i|x_i)}\right)^{\beta}\right) + L_{D_r}(\theta), \tag{51}$$

where π_θ and π_{ref} denote the current model and the reference model before unlearning, respectively, and $\beta > 0$ is an inverse temperature.

D.2 GUARD Objective for Frameworks Other than GA

Given the Forget Set D_f and a model parameterized by θ, with slight abuse of notation, the GUARD losses for additional frameworks are defined as follows.

- GUARD objective under Gradient Difference:

$$L_{GD}^{GUARD}(\theta) := -L_{D_f}^{GUARD}(\theta) + L_{D_r}(\theta)$$

- GUARD objective under KL Minimization:

$$L_{\mathrm{KM}}^{GUARD} = -L_{D_f}^{GUARD}(\theta) + \frac{1}{|D_r|}\sum_{s\in D_r}\frac{1}{|s|}\sum_{i=2}^{|s|}\mathrm{KL}\big(M_0(s_{<i}) \parallel M_{\mathrm{u}}(s_{<i})\big). \tag{52}$$

- GUARD objective under Preference Optimization (PO):

$$L_{\mathrm{PO}}^{GUARD} = L_{D_r}(\theta) + L_{D_f^{\mathrm{idk}}}^{GUARD}(\theta). \tag{53}$$

- GUARD objective under Negative Preference Optimization (NPO):

$$\begin{aligned} L_{\mathrm{NPO}}^{GUARD} &= L_{D_f,\mathrm{NPO}}^{GUARD}(\theta) + L_{D_r}(\theta), \\ \text{where}\quad L_{D_f,\mathrm{NPO}}^{GUARD}(\theta) &:= \frac{1}{n_f}\sum_{i\in I_f}\omega_i^{GUARD}\cdot\frac{2}{\beta}\log\left(1+\left(\frac{\pi_\theta(y_i|x_i)}{\pi_{\mathrm{ref}}(y_i|x_i)}\right)^{\beta}\right). \end{aligned} \tag{54}$$

In all of these objectives, GUARD only reweights the forget-side term, and the retain-side term is unchanged. The first-order analysis of Section 4, which compares a single GA step with a single GUARD step, therefore carries over to the forget-side terms of GD and KM, since their retain-side terms contribute identically to both updates. It also covers the first step of NPO, where all likelihood ratios $\pi_\theta(y_i|x_i)/\pi_{\mathrm{ref}}(y_i|x_i)$ equal one and the gradient of the NPO forget-side term coincides with that of GA. For PO, whose forget-side term is minimized on the neutral responses rather than maximized on the original ones, and for NPO beyond its first step, the reweighting is a heuristic extension that is not covered by the analysis of Section 4.

E Additional results

E.1 Additional Settings

Configuration and Hyperparameters. All experiments were evaluated on equal-cost hardware, Ubuntu 22.04 LTS system with 2 Intel(R) Xeon(R) Gold 6138 CPUs with 20 cores, 4 NVIDIA H100 GPUs with 80 GB memory, and 384 GB RAM. All experiments use 40 CPU cores and 4 GPUs. Unless otherwise specified, the following configurations are used:

`TOFU benchmark:` We *strictly follow* the official protocol of TOFU (Maini et al., 2024) for both fine-tuning and unlearning to ensure that all reported numbers are directly comparable to the official TOFU leaderboard[2]. All baselines, including the loss-reweighting methods and RMU, are run under this single protocol. Since their original papers use different forget splits, training lengths, checkpoint selection, batch sizes, or metrics, our numbers for these methods are not directly comparable with the published ones. Specifically:

- **Optimizer.** AdamW with learning rate 10^{-5} and weight decay 0.01.
- **Schedule.** 5 epochs for both fine-tuning and unlearning, with a linear warm-up during the first epoch.
- **Batch size.** An effective batch size of 32 question–answer pairs on 4 GPUs, matching the effective batch size of TOFU's released configuration.
- **Reporting.** All TOFU results in this paper correspond to the checkpoint at the end of the 5-th unlearning epoch, matching the convention used by the TOFU leaderboard.
- **Data attribution.** We use a temperature $\tau = 0.03$ unless stated otherwise. Attribution scores are computed once on the fine-tuned model checkpoint and reused across epochs.

We deliberately avoid altering any of TOFU's training hyperparameters: tuning epochs, batch size, or learning rate against the unlearning objective could implicitly bias the comparison in favour of GUARD. By inheriting TOFU's setup as-is, the only difference between a baseline and its GUARD-augmented counterpart is the per-sample reweighting introduced by GUARD's attribution scores.

`MUSE benchmark:` We strictly follow the official protocol of MUSE (Shi et al., 2024) so that our numbers for the methods evaluated in the MUSE paper (GA, GA_GDR, GA_KLR, and, up to the choice of β discussed below, NPO_GDR) are directly comparable to the MUSE leaderboard (Shi et al., 2024, Table 3). Specifically:

- **Target models.** We use the official MUSE-released target checkpoints: `LLaMA-2-7B` fine-tuned on BBC NEWS for the NEWS corpus, and `ICLM-7B` fine-tuned on Harry Potter for the BOOKS corpus. These are downloaded directly from the MUSE Huggingface repository.
- **Optimizer.** AdamW with learning rate 10^{-5} and per-device batch size of 32.
- **Schedule.** Up to 10 unlearning epochs with the official MUSE *early-stopping* rule: training is stopped when the unlearned model's utility (Retain Set KnowMem) drops below the released MUSE `Retrain` reference. We report the checkpoint selected by this rule, matching the MUSE leaderboard convention.
- **Method-name mapping.** For consistency with TOFU notation, we use *KM* to refer to MUSE's `GA_KLR` (gradient ascent + KL regularizer on Retain Set), *GD* to refer to MUSE's `GA_GDR` (gradient ascent + gradient descent on Retain Set), and *NPO* to refer to MUSE's `NPO_GDR` (NPO with retain regularizer used in MUSE Table 3). Our GA matches MUSE's pure GA.
- **Data attribution.** We use a temperature $\tau = 0.03$ unless stated otherwise; attribution is computed once on the released target checkpoint and reused across epochs.

As with TOFU, we make no other modifications to the training pipeline beyond GUARD's per-sample reweighting; baseline hyperparameters are kept at their published defaults except where noted below.

[2] `https://huggingface.co/spaces/locuslab/tofu_leaderboard`

Hyperparameter Selection Strategy for GUARD Integration. To evaluate GUARD as a modular and robust enhancement of unlearning schemes, we adopt a conservative experimental design regarding hyperparameter selection. When applying GUARD to a baseline method (e.g., NPO, GA), we intentionally maintained the baseline's hyperparameters (such as learning rate, weight decay, and NPO's β) identical to their standalone settings (for NPO on MUSE, the value $\beta = 0.08$ discussed below, which is shared by NPO and NPO+GUARD). We only introduced GUARD's temperature τ as an additional hyperparameter. This design choice serves two purposes:

1. **Modularity Check:** It ensures that GUARD functions as a "plug-and-play" module that improves performance without requiring extensive method-specific retuning or changing the optimal parameter settings of the underlying algorithm.

2. **Performance Lower Bound:** We note that the hyperparameters optimized for the standalone baseline may not remain the global optimum when combined with GUARD's attribution-driven weighting. Therefore, our reported results likely represent a *lower bound* on the potential performance of GUARD. While jointly optimizing both the baseline's parameters (e.g., NPO's β) and GUARD's τ could yield further improvements, we prioritized demonstrating that GUARD provides substantial gains even under these restrictive, non-co-optimized conditions.

Baseline-Specific Hyperparameters. For all baselines, we keep the algorithm-internal hyperparameters at the values reported in the original papers (or in the official TOFU / MUSE codebases) to ensure that comparisons are not biased by per-method retuning. The two explicit exceptions are **NPO on MUSE**, discussed below, and **RMU**, for which we use the `open-unlearning` configuration. A summary appears in Table 6.

- **GA, PO**: No algorithm-internal hyperparameter beyond the learning rate. The loss is $-L_{D_f}$ (GA) or the equally-weighted sum $L_{D_r} + L_{D_f^{\text{idk}}}$ (PO).

- **GD, KM**: A single retain-regularizer weight, implicitly set to $\lambda = 1$ in TOFU's `dataloader.py` and MUSE's `baselines/iterative.py`, i.e., $L = -L_{D_f} + L_{D_r}$ for GD and $L = -L_{D_f} + \mathrm{KL}(M_0 \,\|\, M_u)$ for KM.

- **NPO** (Zhang et al., 2024a): The inverse-temperature β controls the strength of unlearning: smaller β makes the NPO loss collapse toward GA (more aggressive forgetting), while larger β produces a slower, more retention-preserving divergence. *On TOFU, we use the original paper's default* $\beta = 0.1$. *On MUSE, we deliberately set* $\beta = 0.08$ *(slightly more aggressive than the default).* The reason for this deviation is privacy-driven: with $\beta = 0.1$, NPO leaves substantial residual Forget Set knowledge in the model, and once GUARD's reweighting further protects Retain Set utility, the under-removed forget side translates into a poor PrivLeak score (membership inference becomes easy because the model still assigns high likelihood to forget samples). Lowering β from 0.1 to 0.08 produces a more thorough forget side decay and brings PrivLeak meaningfully closer to the ideal $[-5\%, 5\%]$ window used in the MUSE protocol. This trades a modest amount of Retain KnowMem (utility) for a substantially better privacy outcome, which we judged to be the more meaningful comparison point under the MUSE evaluation.

- **SimNPO** (Fan et al., 2024): $\beta = 4.5$, $\gamma = 0$, `npo_coeff` $= 0.125$, `grad_diff_coeff` $= 1.0$ on TOFU; $\beta = 0.7$ (NEWS) / 0.75 (BOOKS), $\gamma = 0$, `npo_coeff` $= 1.0$, retain weight $= 0.1$ on MUSE. Values follow the SimNPO official TOFU config and MUSE README.

- **FLAT (TV)** (Wang et al., 2024): f-divergence variant set to Total Variation ($f^*(x) = \tanh(x)/2$); `smooth_rate` $= -0.4$; equal weighting $\lambda_e = \lambda_f = 1$ between the template-response and forget-response branches. Values follow the FLAT TOFU config; we apply the same configuration on MUSE.

- **SatImp** (Yang et al., 2025): Token-reweighting exponents $\beta_1 = 5$ (saturation) and $\beta_2 = 1$ (importance); retain regularizer weight $\lambda = 1$. Values follow Section 5 of the original SatImp paper and are reused on MUSE.

- **RMU** (Li et al., 2024b): Steering coefficient $c = 2$, retain-loss weight $\alpha = 1$, target layer $\ell = 7$, forget-loss weight $\gamma = 1$. Values follow the widely-used `open-unlearning` RMU re-tune for TOFU/MUSE; the original WMDP paper reports much larger c and α values that do not transfer well to TOFU/MUSE.
- **GUARD (ours)**: Temperature $\tau = 0.03$ on both TOFU and MUSE, applied identically across GA+GUARD, GD+GUARD, KM+GUARD, PO+GUARD (TOFU only, since PO is omitted on MUSE), and NPO+GUARD. We do not retune the underlying baseline's hyperparameters when integrating GUARD, so the only difference between any baseline and its GUARD-augmented counterpart is the per-sample reweighting derived from a_i^{GUARD}.

Table 6: Algorithm-internal hyperparameters of all baselines and GUARD on TOFU and MUSE. All values follow the corresponding original papers / official codebases except **RMU** and **NPO on MUSE**, where we set $\beta = 0.08$ (vs. the default 0.1) to obtain a more thorough forget-side decay. This brings PrivLeak meaningfully closer to the ideal $[-5\%, 5\%]$ window at the cost of a modest Retain KnowMem decrease. "—" indicates no algorithm-internal hyperparameters present beyond the learning rate.

Method	**Hyperparameter**	**TOFU**	**MUSE**
GA	—	—	—
PO	—	—	—
GD	retain weight λ	1.0	1.0
KM	KL weight λ	1.0	1.0
NPO	inverse temperature β	0.1 (default)	0.08 **(lowered for privacy)**
	retain composite weight	1.0	1.0
SimNPO	β	4.5	$0.7^{(\mathrm{N})}$ / $0.75^{(\mathrm{B})}$
	γ (margin)	0	0
	`npo_coeff` / retain weight	0.125 / 1.0	1.0 / 0.1
FLAT (TV)	`smooth_rate`	-0.4	-0.4
	template / forget weights	1 / 1	1 / 1
SatImp	β_1 / β_2	5 / 1	5 / 1
	retain weight λ	1.0	1.0
RMU	c / α / ℓ / γ	2 / 1 / 7 / 1	2 / 1 / 7 / 1
GUARD (ours)	temperature τ	0.03	0.03

$^{(\mathrm{N})}$ MUSE NEWS; $^{(\mathrm{B})}$ MUSE BOOKS.

E.2 Results for LLaMA-2-7B on the TOFU Data Set

The results are presented in Tables 7 and 8.

E.3 Results for Phi-1.5B on the TOFU Data Set

We present the results of Phi-1.5B in Tables 9, 10 and 11.

Effectiveness of GUARD. Across all data sets, methods, and splits, GUARD consistently reduces the sacrifice rate across key metrics (ρ_r, ρ_p, and ρ_t) wherever the baseline degrades performance (positive values), demonstrating its ability to mitigate unintended knowledge degradation while effectively removing targeted data. The numbers below are reported at the final (5-th) unlearning epoch under TOFU's official protocol; specifically:

- *Forgetting 1% of data:* On the Retain Set, GUARD reduces ρ_r from 4.83 to -1.58 (compared to GA), ρ_p by 13.38% (compared to PO), and ρ_t by 11.99% (compared to PO).
- *Forgetting 5% of data:* On the Retain Set, GUARD lowers ρ_r by 19.54% (compared to KM), ρ_p by 21.65% (compared to KM), and ρ_t by 18.68% (compared to GA).
- *Forgetting 10% of data:* On the Retain Set, GUARD achieves reductions of ρ_r by 19.38% (compared to PO), ρ_p by 30.48% (compared to PO), and ρ_t by 20.64% (compared to GA).

Table 7: Unlearning experiments with LLaMA-2-7B on 1% of all data points in TOFU under the official TOFU protocol; all values are reported at the final (5-th) unlearning epoch. GUARD consistently reduces the Sacrifice Rates across all baselines and evaluation metrics wherever the baseline degrades performance (positive values).

Methods	**Retain Set(↓)**			**Real Author Set(↓)**			**Real World Set(↓)**		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
GA	17.53	3.94	**-2.81**	5.56	4.22	52.92	0.44	1.86	1.94
GA + GUARD	**12.30**	**2.75**	-2.00	**4.00**	**3.10**	**38.00**	**0.30**	**1.30**	**1.50**
KM	18.28	3.90	**-3.04**	5.72	4.37	56.30	**-1.33**	1.90	3.26
KM + GUARD	**12.50**	**2.80**	-2.20	**4.20**	**3.20**	**40.00**	-0.90	**1.40**	**2.50**
PO	11.20	16.73	29.90	**-1.48**	**-8.36**	**-12.06**	0.39	**-21.10**	**-7.41**
PO + GUARD	**8.50**	**12.50**	**22.00**	-1.20	-6.50	-9.00	**0.30**	-16.00	-5.50
GD	16.67	3.84	**-5.99**	7.26	4.99	73.96	0.46	2.16	**-6.25**
GD + GUARD	**12.00**	**2.80**	-4.20	**5.50**	**3.80**	**55.00**	**0.30**	**1.50**	-4.50

Table 8: Unlearning experiments with LLaMA-2-7B on 5% of all data points in TOFU under the official TOFU protocol; all values are reported at the final (5-th) unlearning epoch. GUARD consistently reduces the Sacrifice Rates across all evaluation metrics and baselines wherever the baseline degrades performance (positive values), with the most substantial gains observed on the Retain Set for GD (ρ_r and ρ_p reduced by 39.32% and 47.14%) and PO (ρ_t reduced by 39.53%).

Methods	**Retain Set(↓)**			**Real Author Set(↓)**			**Real World Set(↓)**		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
GA	100.14	99.63	204.51	94.56	17.50	103.89	89.64	16.38	113.42
GA + GUARD	**83.31**	**86.52**	**158.75**	**77.47**	**13.61**	**84.57**	**72.03**	**12.14**	**96.46**
KM	99.68	100.02	454.89	95.09	18.71	281.02	89.10	12.69	212.39
KM + GUARD	**82.16**	**86.39**	**418.08**	**76.11**	**14.93**	**261.36**	**71.02**	**8.38**	**195.91**
PO	33.52	41.04	66.15	78.82	26.54	121.14	14.30	9.16	78.52
PO + GUARD	**20.48**	**28.83**	**40.00**	**68.21**	**18.62**	**101.05**	**12.83**	**5.14**	**64.33**
GD	90.51	83.92	**-58.70**	54.91	4.92	21.01	12.91	**-0.59**	**-9.38**
GD + GUARD	**54.92**	**44.36**	-18.88	**32.40**	**3.50**	**12.50**	**8.30**	-0.40	-6.10

Data Set-Specific Observations. Across all data sets, splits, and evaluation metrics, GUARD reduces every Retain-Set ρ_r and ρ_p, which aligns with its design: the proxy data attribution is computed based on the alignment between the forget data and the overall retain data, directly benefiting retention performance. In contrast, improvements on the generalization sets (Real World Set and Real Author Set) are indirect. Since unlearning the Forget Set inherently reduces overfitting, GUARD in some cases turns a positive sacrifice rate of the baseline on the Real World Set or the Real Author Set into a negative one (e.g., ρ_p of GA on the Real World Set at 5%, from 2.46 to -0.97), which suggests that GUARD not only prevents unintended knowledge degradation but may also enhance model generalization.

E.4 Results for ICLM-7B on the MUSE BOOKS Data Set

We present the results of ICLM-7B on MUSE BOOKS data set in Table 12. As on MUSE NEWS, GUARD improves Retain KnowMem for every baseline at comparable Forget Set VerbMem and KnowMem, and NPO+GUARD attains the highest Retain KnowMem (52.4).

Table 9: Sacrifice rate (%) of different unlearning methods when unlearning Phi-1.5B on 1% of TOFU data, under the official TOFU protocol; all values are reported at the final (5-th) unlearning epoch. Across all datasets (Retain Set, Real Author Set, and Real World Set), GUARD consistently reduces the sacrifice rate across various methods (GA, GD, KM, PO) wherever the baseline degrades performance (positive values). On the Retain Set, GUARD lowers ρ_r of GA from 4.83 to -1.58, ρ_p by 13.38% over PO, and ρ_t by 11.99% over PO.

Methods	**Retain Set**(↓)			**Real Author Set**(↓)			**Real World Set**(↓)		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
GA	4.83	2.27	**-0.35**	**-17.79**	3.01	117.87	7.01	2.97	105.41
GA+GUARD	**-1.58**	**0.85**	0.77	-15.90	**1.61**	**104.06**	**6.17**	**2.15**	**104.72**
GD	9.21	0.96	210.08	**-6.35**	0.89	**-63.85**	10.91	0.44	32.37
GD+GUARD	**4.77**	**-0.20**	**208.53**	-4.23	**-0.43**	-46.32	**10.64**	**-0.31**	**30.40**
KM	5.82	2.59	**-8.36**	**-9.09**	1.85	114.83	10.47	2.23	112.09
KM+GUARD	**0.51**	**1.56**	-7.41	-7.18	**0.59**	**99.71**	**10.01**	**1.85**	**111.17**
PO	27.87	30.57	65.98	54.31	**-17.04**	**-27.08**	46.64	**-22.50**	**-89.73**
PO+GUARD	**25.06**	**26.48**	**58.07**	**53.98**	-14.67	-23.97	**46.49**	-17.14	-88.02

E.5 Computational Efficiency Analysis

To evaluate the computational overhead introduced by GUARD, we conducted runtime comparisons between standard Gradient Ascent (GA) and GA augmented with GUARD across different data sets and model architectures. Table 13 presents the per-epoch runtime measurements under TOFU's official protocol (effective batch size 32) and the MUSE protocol on our experimental setup with 4 NVIDIA H100 GPUs. Each epoch time includes the end-of-epoch evaluation. The GA + GUARD column reports its first epoch, which additionally computes the attribution scores, and later epochs reuse the cached scores and cost the same as the baseline.

We point out that the runtime reported for GA + GUARD represents the **total integrated time**, encompassing both the unlearning optimization and the necessary gradient computations for data attribution. As shown in Table 13, GUARD introduces minimal overhead (approximately 11.5% of one baseline epoch on average, with the per-setting range $\sim$ 7–21%, incurred only in the first epoch). For instance, on the TOFU 10% split with LLaMA-2-7B, GUARD adds only 3.2 minutes to the 32.3-minute baseline (in the first epoch only). This efficiency stems from two key factors:

1. **Efficient Gradient Computation:** The per-sample gradient computation (as defined in Eq. (2)) requires only standard forward-backward passes. Unlike conventional influence functions, GUARD avoids computationally prohibitive Hessian estimation or model retraining.

2. **Amortization via Preprocessing:** The attribution score calculation serves as a one-time preprocessing step. Once computed for a given model-data set combination, these scores are cached and reused across multiple unlearning epochs or hyperparameter tuning experiments, significantly amortizing the initial cost.

In conclusion, the modest runtime increases observed across diverse experimental settings confirm that GUARD is a computationally viable enhancement. By avoiding the bottleneck of repetitive gradient recalculations and Hessian approximations, GUARD achieves substantial retention improvements without imposing prohibitive computational costs.

E.6 Robustness with Respect to Limited and Synthetic Retain Sets on MUSE

We apply the same robustness protocol described in the main paper (Section 5, Table 5) to MUSE NEWS (LLaMA-2-7B) and MUSE BOOKS (ICLM-7B). All other hyperparameters (NPO's β, GUARD's temperature τ) are kept at the values used in the main MUSE experiments (Tables 4 and 12); only the gradient $J_{\text{avg}}^{D_r}$ used to compute the attribution score is replaced by its 20%-subsample or GPT-4 synthetic counterpart.

Table 10: Sacrifice Rate of GUARD upon unlearning Phi-1.5B on 5% of all datapoints in TOFU, under the official TOFU protocol; all values are reported at the final (5-th) unlearning epoch. When unlearning a larger portion (compared to the portion of 1% in Table 9) of the dataset, GUARD generally induces a larger absolute reduction on the sacrifice rate across all evaluation sets (Retain Set, Real Author Set, and Real World Set) and methods (GA, GD, KM, and PO). On the Retain Set, GUARD lowers ρ_r by 19.54% over KM, ρ_p by 21.65% over KM, and ρ_t by 18.68% over GA. On the Real World Set, GA+*GUARD* achieves a negative ρ_p while GA does not, indicating that GUARD can in some cases improve generalization by reducing overfitting on the Forget Set.

Methods	**Retain Set**(↓)			**Real Author Set**(↓)			**Real World Set**(↓)		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
GA	96.88	95.89	102.48	34.08	2.14	36.04	23.06	2.46	22.87
GA+GUARD	**78.49**	**79.21**	**83.34**	**25.77**	**-0.20**	**31.01**	**7.87**	**-0.97**	**11.20**
GD	81.30	46.58	40.28	35.73	**-0.61**	10.57	23.39	1.50	28.14
GD+GUARD	**67.92**	**34.43**	**36.41**	**26.66**	-0.53	**8.02**	**15.11**	**1.33**	**24.53**
KM	96.65	94.02	94.36	29.07	2.61	37.23	33.37	3.69	31.62
KM+GUARD	**77.76**	**73.66**	**76.23**	**20.46**	**0.22**	**31.83**	**18.61**	**0.04**	**21.29**
PO	47.22	53.08	106.86	14.21	9.25	103.74	4.35	-0.09	**-21.03**
PO+GUARD	**39.11**	**41.39**	**88.70**	**8.23**	**5.57**	**82.02**	**3.11**	**-0.40**	-18.63

Table 14 reports the four MUSE metrics across NPO, NPO+GUARD (full / 20% / synthetic retain), and RMU at full-retain access (included as a reference for the strongest competing baseline). Mirroring the TOFU pattern, both GUARD ablations preserve most of full-retain GUARD's Retain KnowMem gain over NPO on both NEWS and BOOKS: on NEWS the Retain KnowMem moves from 37.2 (NPO) to 47.6 (20%) and 46.5 (synthetic), close to the 50.0 obtained with full-retain; on BOOKS it moves from 24.3 (NPO) to 49.0 (20%) and 47.0 (synthetic), close to the full-retain 52.4. Notably, on NEWS the synthetic-retain variant (46.5) already exceeds RMU at full-retain access (43.1); on BOOKS it is comparable to RMU at full-retain (47.0 vs. 47.8), despite using *no* original retain data in the attribution. Forget-side metrics (VerbMem, KnowMem) remain essentially unchanged across all settings, confirming that surrogate-based attribution does not compromise forgetting effectiveness. Interestingly, PrivLeak moves slightly closer to 0 as the retain surrogate becomes more approximate ($98.2 \to 97.5 \to 96.2$ on NEWS; $-38.4 \to -37.1 \to -36.2$ on BOOKS), suggesting that less precise retain matching also incidentally relaxes membership-inference exposure—a small privacy benefit that partially offsets the slight Retain KnowMem drop relative to full-retain.

These MUSE findings, taken together with the TOFU results in the main paper, validate GUARD's practical robustness when the original Retain Set is partially or entirely inaccessible. For surrogate-construction strategies beyond random subsampling and LLM generation, prior work on Fisher-information sampling (Liu et al., 2024b) and coreset methods (Mirzasoleiman et al., 2020) can be plugged into GUARD without modification.

E.7 The Synthetic Retain Set: Pipeline and Quality Validation

The synthetic-retain ablation reported in the main paper (Table 5) and in Appendix E.6 (Table 14) addresses the strictly harder case in which *no* portion of the original Retain Set is accessible to GUARD's attribution—e.g., when the data has been deleted for compliance reasons, or when an organization deploys an unlearning pipeline on a third-party model whose training corpus is unknown. In such settings, GUARD's attribution score $a_i^{\text{GUARD}} = [J_{\text{avg}}^{D_r}]^\top J_i(\theta_0)$ is no longer directly computable, since $J_{\text{avg}}^{D_r}$ requires gradient evaluation on the retain corpus. Our workaround is to replace D_r with a *synthetic retain proxy* $\tilde{D}_r$ produced by an external LLM (here, GPT-4) prompted to generate samples that share only the general format and topic of the natural retain corpus of each benchmark, using only public benchmark information that typically survives compliance-driven data deletion; the attribution score becomes $a_i^{\text{GUARD-syn}} = [J_{\text{avg}}^{\tilde{D}_r}]^\top J_i(\theta_0)$ and plugs into the standard GUARD pipeline without modification. Note that the synthetic proxy replaces D_r only in GUARD's attribution. Base methods whose own objective contains a retain term (GD, KM, PO, and NPO)

Table 11: Sacrifice Rate of GUARD upon unlearning Phi-1.5B on 10% of all datapoints in TOFU, under the official TOFU protocol; all values are reported at the final (5-th) unlearning epoch. When unlearning a larger portion of the dataset, GUARD generally induces an even larger absolute reduction on the sacrifice rate across all evaluation sets (Retain Set, Real Author Set, and Real World Set) and methods (GA, GD, KM, and PO). On the Retain Set, GUARD lowers ρ_r by 19.38% over PO, ρ_p by 30.48% over PO, and ρ_t by 20.64% over GA.

Methods	**Retain Set**(↓)			**Real Author Set**(↓)			**Real World Set**(↓)		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
GA	100.25	99.77	124.13	47.76	10.96	28.72	86.85	17.56	61.66
GA+GUARD	**81.50**	**83.43**	**98.51**	**38.20**	**8.50**	**22.10**	**67.30**	**12.40**	**42.80**
GD	80.69	68.66	22.79	44.41	**-0.62**	**-12.76**	37.96	0.78	**-16.15**
GD+GUARD	**63.20**	**52.40**	**16.80**	**32.50**	-0.50	-9.40	**27.40**	**0.55**	-11.50
KM	100.59	99.77	121.35	47.44	11.67	27.42	86.27	17.81	59.41
KM+GUARD	**82.10**	**83.50**	**96.40**	**38.50**	**9.30**	**21.20**	**67.80**	**12.80**	**41.30**
PO	35.76	45.24	65.17	21.64	13.91	180.20	4.19	**-1.04**	8.43
PO+GUARD	**28.83**	**31.45**	**43.80**	**14.50**	**9.20**	**152.30**	**2.80**	-0.70	**5.60**

Table 12: Unlearning experiments with ICLM-7B on the MUSE BOOKS dataset under the official MUSE protocol; "w/o Unlearn" is the released MUSE BOOKS target model.

Methods	**Forget Set VerbMem** (↓)	**Forget Set KnowMem** (↓)	**PrivLeak** ($\in [-5\%, 5\%]$)	**Retain Set KnowMem** (↑)
		BOOKS		
w/o Unlearn	99.8	59.4	-57.5	66.9
GA	**0.0**	**0.0**	**-25.4**	0.0
GA + GUARD	**0.0**	**0.0**	-31.0	**10.6**
KM	**15.7**	22.3	**-39.5**	36.2
KM + GUARD	16.2	**20.4**	-41.8	**47.5**
GD	**0.0**	**0.0**	-28.8	11.4
GD + GUARD	**0.0**	**0.0**	**-26.9**	**27.9**
NPO	**0.0**	**0.0**	-32.0	24.3
SimNPO	**0.0**	**0.0**	-34.8	47.5
FLAT (TV)	**0.0**	**0.0**	-45.4	28.7
SatImp	**0.0**	**0.0**	**-26.6**	22.7
RMU	**0.0**	**0.0**	-36.1	47.8
NPO + GUARD	**0.0**	**0.0**	-38.4	**52.4**

still use D_r for training, so the whole pipeline is free of the original retain data only for base methods without such a term, such as GA.

Synthesis Pipelines. For each benchmark we use a single, lightweight prompt:

- *TOFU.* We generate 200 *new* fictitious authors (each with a JSON profile of biographical metadata) and 20 question–answer pairs per author, yielding $\sim$ 4,000 synthetic QA pairs that match the scale of the full TOFU dataset. Author names are explicitly disallowed to overlap with any Forget Set author.
- *MUSE NEWS.* We generate $\sim$ 1.6M tokens of BBC-style news articles spanning sports, weather, business, and arts—topics chosen to be disjoint from the forget split's political-news topics.
- *MUSE BOOKS.* We generate $\sim$ 0.5M tokens of fantasy fiction excerpts featuring magical schools / wizards, with the prompt explicitly forbidding any Harry Potter character or place names.

The total GPT-4 API cost across all three settings is approximately $40.

Table 13: Per-epoch runtime of unlearning with and without GUARD, measured under the official TOFU and MUSE protocols. Attribution is the difference between the two runtimes, and Overhead is Attribution relative to the GA runtime.

Dataset	Model	Forget set	GA (min)	GA + GUARD (min)	Attribution (min)	Overhead
TOFU	LLaMA-2-7B	1%	4.2	5.1	0.9	21.4%
		5%	14.4	16.6	2.2	15.3%
		10%	32.3	35.5	3.2	9.9%
	Phi-1.5B	1%	2.8	3.0	0.2	7.1%
		5%	6.1	6.8	0.7	11.5%
		10%	14.5	15.6	1.1	7.6%
MUSE	LLaMA-2-7B	NEWS	83.7	91.2	7.5	9.0%
	ICLM-7B	BOOKS	116.0	127.9	11.9	10.3%

Quality Validation. To verify that $\tilde{D}_r$ does not leak Forget Set content, we compute (i) the maximum BERT-embedding cosine similarity between any synthetic sample and any Forget Set sample (max sim. ≤ 0.68 across all three benchmarks, well below the ≥ 0.92 typical of paraphrase pairs); and (ii) GPT-4-as-judge labels less than 3% of synthetic samples as containing Forget Set entities. The synthetic data therefore shares the format and topic of the natural retain corpus while remaining nearly content-disjoint from the Forget Set.

Table 14: Robustness of GUARD under limited (random 20%) and synthetic (GPT-4 generated, with no original retain data used in the attribution) Retain Sets on MUSE NEWS (LLaMA-2-7B) and MUSE BOOKS (ICLM-7B), at the early-stopping checkpoint under MUSE's official protocol. RMU is included at full retain access as a reference point for the strongest competing baseline. As on TOFU (Table 5), both GUARD ablations preserve most of full-retain GUARD's Retain KnowMem gain over NPO; on NEWS, the synthetic-retain variant already outperforms RMU at full retain access, and on BOOKS it remains comparable. Bold marks the column-wise best across all rows; PrivLeak is reported for completeness (values closer to 0 are preferred).

Methods	MUSE NEWS (LLaMA-2-7B)				MUSE BOOKS (ICLM-7B)			
	Forget VerbMem (↓)	Forget KnowMem (↓)	PrivLeak (∈ [−5%, 5%])	Retain KnowMem (↑)	Forget VerbMem (↓)	Forget KnowMem (↓)	PrivLeak (∈ [−5%, 5%])	Retain KnowMem (↑)
NPO	**1.5**	44.8	**93.0**	37.2	0.0	0.0	**-32.0**	24.3
RMU (full retain, reference)	**1.5**	**44.0**	95.0	43.1	0.0	0.0	-36.1	47.8
NPO + GUARD (full retain)	1.6	45.2	98.2	**50.0**	0.0	0.0	-38.4	**52.4**
NPO + GUARD (20% retain)	1.6	45.4	97.5	47.6	0.0	0.0	-37.1	49.0
NPO + GUARD (synthetic retain)	1.6	45.6	96.2	46.5	0.0	0.0	-36.2	47.0

E.8 Proxy for Data Attribution

For the unification methods, the reversely unified scores controlled by temperature are plotted in Fig. 4. Exponential unification provides a smoother attribution distribution at a higher temperature.

E.9 Detailed Results for Alternative Attribution Methods

In Section 5, we compared GUARD with three alternative attribution mechanisms: Embedding Similarity (ES), RelatIF, and RIF. Here we provide the detailed quantitative results. Table 15 presents the Sacrifice Rates, and Table 16 presents the Absolute Performance metrics. Consistent with our main findings, GUARD achieves superior retention performance (lower sacrifice rates ρ_r and ρ_p and higher Retain ROUGE-L and Probability) compared to these first-order approximations and feature-space methods. Here, the Embedding Similarity (ES) score of a forget sample is the cosine similarity between its BERT embedding and the average BERT embedding of the Retain Set. The first-order RelatIF score is $a_i^{\mathrm{GUARD}}/\|J_i(\theta_0)\|_2$, i.e., the RelatIF normalization with $H \approx I$, and the first-order RIF score is $(1-h_i)^{-1}a_i^{\mathrm{GUARD}}$, where the generalized leverage score h_i is also evaluated with $H \approx I$.

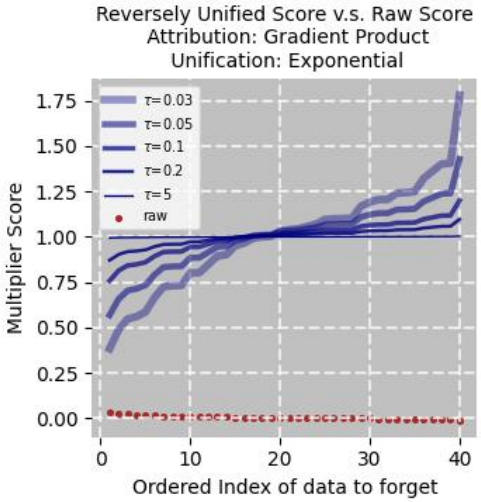


Figure 4: Raw Score and Inverse unified score under different temperatures τ for the 40 samples of the TOFU forget01 split (Phi-1.5B), ordered by score (gradient-product attribution, exponential unification).

Table 15: Comparison of the performance of different attribution methods integrated with NPO on the TOFU benchmark (LLaMA-2-7B, 10% forget split) under the official TOFU protocol; values reported at the final (5-th) unlearning epoch. GUARD outperforms both the feature-based baseline (ES) and other first-order influence approximations (RelatIF, RIF) across all evaluation sets in ρ_r and ρ_p. The substantially lower sacrifice rates on the Retain Set demonstrate that the gradient-alignment strategy of GUARD minimizes unintended harm to preserved knowledge more effectively than alternative attribution metrics.

Methods	**Retain Set(↓)**			**Real Author Set(↓)**			**Real World Set(↓)**		
	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t	ρ_r	ρ_p	ρ_t
NPO + ES	26.53	9.04	**-12.54**	19.60	10.20	**-17.40**	4.80	4.10	**-14.30**
NPO + RelatIF	22.08	6.35	-9.85	17.40	9.10	-14.50	4.10	3.50	-13.10
NPO + RIF	19.08	4.52	-8.50	16.20	8.70	-13.20	3.60	3.20	-12.30
NPO + GUARD	**14.32**	**2.87**	-6.10	**13.59**	**7.14**	-10.44	**2.70**	**2.23**	-10.29

F Future Work

We see several directions for future work. (i) *Adaptive attribution schedules*: re-computing or smoothly updating the proxy attribution every few unlearning steps, possibly via a low-rank gradient sketch, could close the gap with a fully online influence-function while keeping the cost amortized. (ii) *Method-agnostic theory*: extending the Sacrifice-Rate analysis beyond GA to a general family of unlearning objectives—including bounded-loss methods (NPO, SimNPO), regularized variants (KL/GD), and feature-space methods (RMU)—would clarify when GUARD's reweighting is provably tight. (iii) *Beyond entity-level unlearning*: applying GUARD to behavior-level alignment, copyrighted-content removal at scale, and sequential "Right-to-be-Forgotten" streams would test the framework's practical reach; combining GUARD with certified unlearning bounds is a particularly promising direction. (iv) *Better synthetic retain proxies*: beyond GPT-4 generation, exploring retrieval-and-paraphrase pipelines, self-consistency-filtered model self-generation, and embedding-space coverage objectives could reduce the gap between synthetic and full-retain GUARD on the harder TOFU split. We view these directions as natural next steps for retention-aware unlearning at deployment scale.

Table 16: Comparison of absolute performance metrics between GUARD and alternative attribution methods integrated with NPO, under the official TOFU protocol; values reported at the final (5-th) unlearning epoch. Per-metric arrows follow TOFU (Maini et al., 2024). While all methods achieve comparable forgetting strength on the Forget Set, GUARD delivers the highest Retain ROUGE-L and Probability.

Methods	**Retain Set**			**Forget Set**		
	ROUGE-L (↑)	Prob. (%) (↑)	T.Ratio (↑)	ROUGE-L (↓)	Prob. (%) (↓)	T.Ratio (↑)
NPO + ES	81.50	93.50	**49.20**	34.20	38.80	**65.40**
NPO + RelatIF	84.30	95.10	48.80	33.90	38.50	65.10
NPO + RIF	86.20	96.20	48.60	**33.70**	38.30	64.90
NPO + GUARD	**89.30**	**97.20**	48.30	33.80	**38.20**	65.20